

Triage Classifier: 10,000-Record Study

Complete comparison using the latest uploaded classifier and the newly supplied dataset

The study retains the uploaded classifier's Logistic Regression, Hist Gradient Boosting and SapBERT baseline settings. It adds Random Forest, alternative text encoders, fold-local preprocessing, dimensionality comparisons, class balancing, feature tests and development-only ensemble selection. Existing application models were not replaced.

Data: 10,000 records; 8,001 development and 1,999 final test records. The target has three labels: 1, 2 and 3. There are no level-4 examples in this file.

Selected system: final held-out results

Metric	Value	95% group-bootstrap interval
Accuracy	99.05%	98.62% - 99.42%
Macro F1	98.88%	98.38% - 99.35%
Macro precision	99.01%	-
Macro recall	98.76%	-

Selection was frozen using development predictions before evaluating the final test. Its components are:

Weight 1.000: Hist Gradient Boosting; SapBERT concepts + patient features / PCA-128 / derived features.

Class-probability multipliers for labels 1, 2 and 3: [0.8, 1.0, 0.8]. These decision adjustments were tuned on development out-of-fold predictions only; adjusted scores are not calibrated probabilities.

How to interpret the scores

This dataset is different from the earlier 821-record, four-class study. A higher score here cannot be attributed to model tuning alone. The user confirmed that the labels were assigned by an AI model. Scores measure agreement with those AI-generated labels, not clinician agreement. No clinician validation or real-patient provenance was provided.

Strong ECG and vital-sign relationships with the labels are documented later in this report. They can make this particular dataset easier to classify. Development CV is used for selection and is not an unbiased final performance estimate.

Outcome of tuning

The frozen selected system did not outperform the source-setting Hist Gradient Boosting baseline on this final test (baseline accuracy 99.15%, macro F1 0.9901 with fusion). Random Forest recorded the highest observed comparison accuracy, 99.35%, and macro F1 0.9926. Selection is not changed after examining test results. These findings do not support a claim that every tuning step improved the strongest baseline.

Previous and current experiments: overview

This table keeps the study settings separate. Different datasets, target classes, encoders and split methods mean that the rows are not a controlled before/after comparison.

Study	Records	Classes	Test rows	Accuracy*	Macro F1*
Earlier SBERT	2,252	4	451	82.93%	78.68%
Earlier SapBERT	821	4	212	76.42%	68.60%
Current selected system	10,000	3	1999	99.05%	98.88%

*For historical studies, accuracy and macro F1 are the highest values separately reported and may come from different classifiers. The current row reports both metrics for the single system frozen before final-test evaluation.

What each study tested

Earlier SBERT: all-mpnet-base-v2 embeddings of English translations; 768-D versus PCA-64; text alone and fusion; three classifiers. A stratified random row split was used, so repeated complaint phrases could occur across the split.

Earlier SapBERT: 821 repository records with matched clinical concepts; 609/212 concept-grouped holdout; source Logistic Regression and Hist Gradient Boosting, with added 768-D and Random Forest comparisons. Its reported CV used preprocessing fitted before the CV folds and was exploratory.

Current study: 10,000 supplied records with user-confirmed AI-assigned labels; three frozen encoders; grouped three-fold screening and five-fold refinement; preprocessing fitted within folds; separate frozen final test; sensitivity checks, label-shuffle control, learning curves and uncertainty intervals.

Complete historical reports

The original SBERT comparison and the expanded SapBERT comparison are included unchanged as appendices after the new study. Their earlier page numbers and values are preserved. The expanded SapBERT report contains the original six configurations plus the additional comparisons and class-level results.

The new experiment tables report measured values only. Any further table supplied by the user can be added without rerunning or retuning the final test.

Dataset preparation and audit

Item	Result
Uploaded size	10,000 rows x 19 columns
Retained size	10,000 rows x 13 columns
Exact duplicate rows removed	0
Normalized text groups	6573
Largest group	30
Shared development/test groups	0

Features retained

Age; gender; arrival mode; original Roman Urdu complaint; generated clinical concept; heart rate; systolic and diastolic blood pressure; ECG status; temperature; oxygen saturation; AVPU. Labels is the target and never a predictor.

Columns removed

Category (constant Cardiac), fuzzy_glossary, min_confidence, unresolved_tokens, seconds and examples_used. These are a constant category or processing metadata, not inputs used to select triage severity.

One arrival-mode value, roman_urdu_reference at source CSV row 21, was set to missing. The row and its label were retained. Missing-value imputation is learned within each training fold. No difficult examples or label classes were removed to improve scores.

Label	Entire dataset	Development	Final test
1	3389	2712	677
2	4716	3773	943
3	1895	1516	379

Source checksum (SHA-256): f9fc4398820661b6005e66c7efd65fd5753c9c92f0ef98a2c37ee87785a27ec2

Evaluation and optimization methods

1. Frozen grouped holdout

Normalize case, punctuation and whitespace in both text columns. Connected components link rows sharing either normalized complaint or concept. A seeded stratified grouped split reserves approximately 20% as the final test set. This prevents exact normalized text overlap; it does not guarantee independence of semantically similar generated templates.

2. Frozen text encoders

SapBERT uses CLS pooling on clinical concepts (768 dimensions, 64-token limit). MPNet uses its Sentence Transformer pooling on concepts (768 dimensions, 128-token limit). Multilingual MiniLM embeds the original Roman Urdu complaints (384 dimensions, 128-token limit). All embeddings are L2-normalized and computed locally with frozen cached checkpoints.

3. Training-only preprocessing

Numeric median imputation and standardization, categorical imputation and one-hot encoding, and PCA are fitted inside each CV training fold. PCA sizes 32, 64 and 128, plus full embeddings, are compared. Source baselines retain the original automatic PCA solver; exploratory candidates explicitly use a randomized solver.

4. Model and feature search

Logistic Regression regularization, tree settings, balanced versus unweighted training, PCA size and text contribution weight are compared. Optional deterministic features are systolic minus diastolic pressure, heart-rate/systolic-pressure ratio and $(\text{systolic} + 2 \times \text{diastolic})/3$. They are transformations of existing inputs, not new measured clinical variables.

5. Grouped cross-validation

Three-fold grouped screening ranks candidates. Five-fold grouped refinement evaluates the leading configurations and leading classifier/input families. Configurations and per-fold scores are saved. The additional ECG/arrival/gender sensitivity tests and learning curves use development rows only.

6. Ensemble and decision adjustment

Compare leading single models and probability averages of classifier-family winners. Small class-probability adjustments are selected on five-fold out-of-fold development predictions, prioritizing macro F1 and then accuracy. This tuning makes those development selection scores optimistic; the separate final test provides the reported external-to-selection check.

7. Final evaluation

Freeze model members, weights and decision adjustments. Refit on all development records, evaluate the final test once, and report macro/per-class metrics and confusion matrices. Group bootstrap with 1,000 replicates gives uncertainty intervals for the selected system. Baselines and CV-family winners are comparisons, not candidates selected by final test scores.

Final held-out comparison

All rows below use the same 1999 final test records. B = source-setting baseline; C = classifier-family winner selected by CV; S = final selected system. Precision, recall and F1 are macro-averaged.

Ref.	Accuracy	Precision	Recall	F1	QWK	MAE
C1	99.10%	99.05%	98.85%	98.95%	0.991	0.009
C2	98.55%	98.46%	98.35%	98.41%	0.986	0.015
C3	99.35%	99.04%	99.50%	99.26%	0.994	0.007
B4	98.60%	98.07%	98.85%	98.44%	0.986	0.014
B5	99.15%	99.32%	98.78%	99.04%	0.992	0.009
B6	59.88%	64.10%	58.88%	60.83%	0.485	0.430
B7	59.93%	66.70%	57.05%	59.65%	0.467	0.419
B8	98.50%	98.03%	98.66%	98.34%	0.985	0.015
B9	99.15%	99.24%	98.78%	99.01%	0.992	0.009
S10	99.05%	99.01%	98.76%	98.88%	0.991	0.010

Model key

C1: Hist Gradient Boosting; SapBERT concepts + patient features / PCA-128 / derived features; 153 predictors.

C2: Logistic Regression; SapBERT concepts + patient features / full dimensions; 790 predictors.

C3: Random Forest; SapBERT concepts + patient features / PCA-64; 86 predictors.

B4: Logistic Regression; Patient features; 22 predictors.

B5: Hist Gradient Boosting; Patient features; 22 predictors.

B6: Logistic Regression; SapBERT concepts / PCA-64; 64 predictors.

B7: Hist Gradient Boosting; SapBERT concepts / PCA-64; 64 predictors.

B8: Logistic Regression; SapBERT concepts + patient features / PCA-64; 86 predictors.

B9: Hist Gradient Boosting; SapBERT concepts + patient features / PCA-64; 86 predictors.

S10: Selected system, specified on page 1.

A baseline score here is measured on this new dataset and grouped holdout. The earlier study's 73-76% values are not valid denominators for attributing improvement to these changes.

Performance comparison of ED triage models

Article-style table matching the supplied screenshot: Model, Accuracy, Recall and F1. Values below are measured on the current frozen final test set and shown on a 0-1 scale. Recall and F1 are macro-averaged.

Model	Accuracy	Recall	F1
C1: Hist Gradient Boosting; SapBERT concepts + patient features / PCA-128 / derived features	0.991	0.988	0.989
C2: Logistic Regression; SapBERT concepts + patient features / full dimensions	0.985	0.984	0.984
C3: Random Forest; SapBERT concepts + patient features / PCA-64	0.993	0.995	0.993
B4: Logistic Regression; Patient features	0.986	0.988	0.984
B5: Hist Gradient Boosting; Patient features	0.991	0.988	0.990
B6: Logistic Regression; SapBERT concepts / PCA-64	0.599	0.589	0.608
B7: Hist Gradient Boosting; SapBERT concepts / PCA-64	0.599	0.570	0.596
B8: Logistic Regression; SapBERT concepts + patient features / PCA-64	0.985	0.987	0.983
B9: Hist Gradient Boosting; SapBERT concepts + patient features / PCA-64	0.991	0.988	0.990
Proposed system (selected on development data)	0.990	0.988	0.989

The proposed-system row uses the actual frozen selection, which may be a single model or an ensemble. Its name is not forced to SapBERT if another configuration was selected. No published comparator is measured on this same test set.

Published context and current result

Performance comparison in the requested Model / Accuracy / Recall / F1 format. Literature rows are reproduced from the cited primary study; the final row is measured in this experiment.

Model	Accuracy	Recall	F1
TF-IDF + Logistic Regression [L1]	0.750	0.988	0.544
TF-IDF + Random Forest [L1]	0.751	0.964	0.582
BiLSTM [L1]	0.746	0.846	0.670
CNN [L1]	0.723	0.787	0.667
Proposed system (current study)	0.990	0.988	0.989

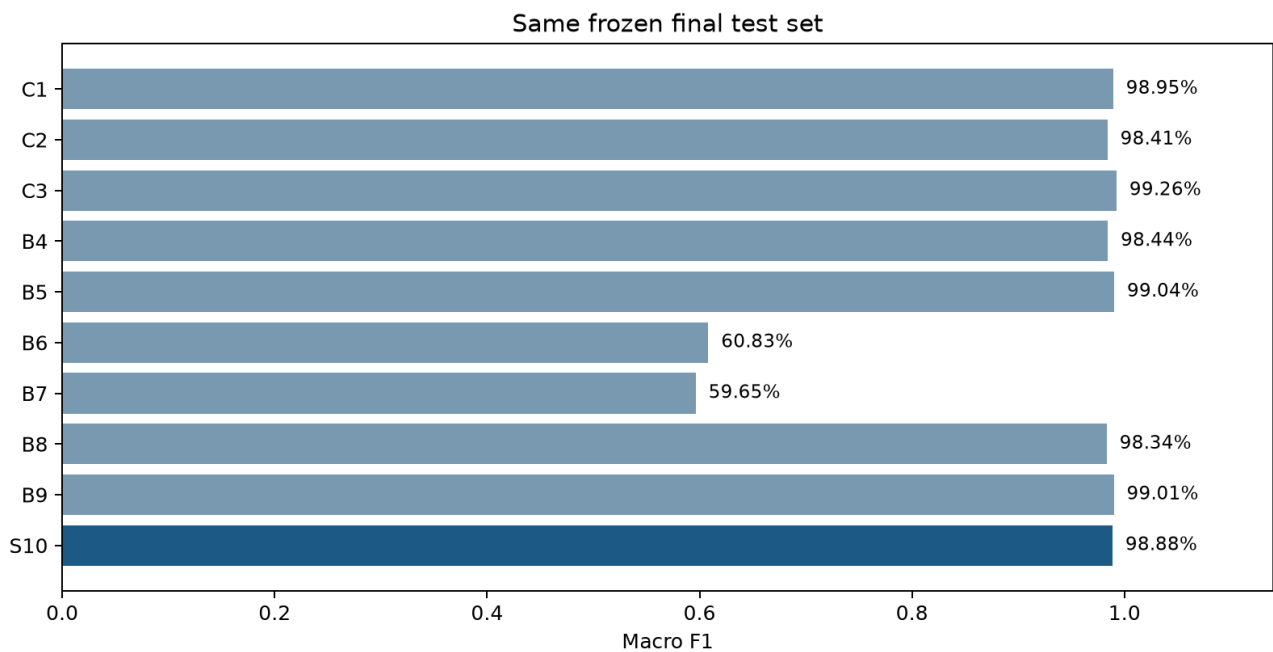
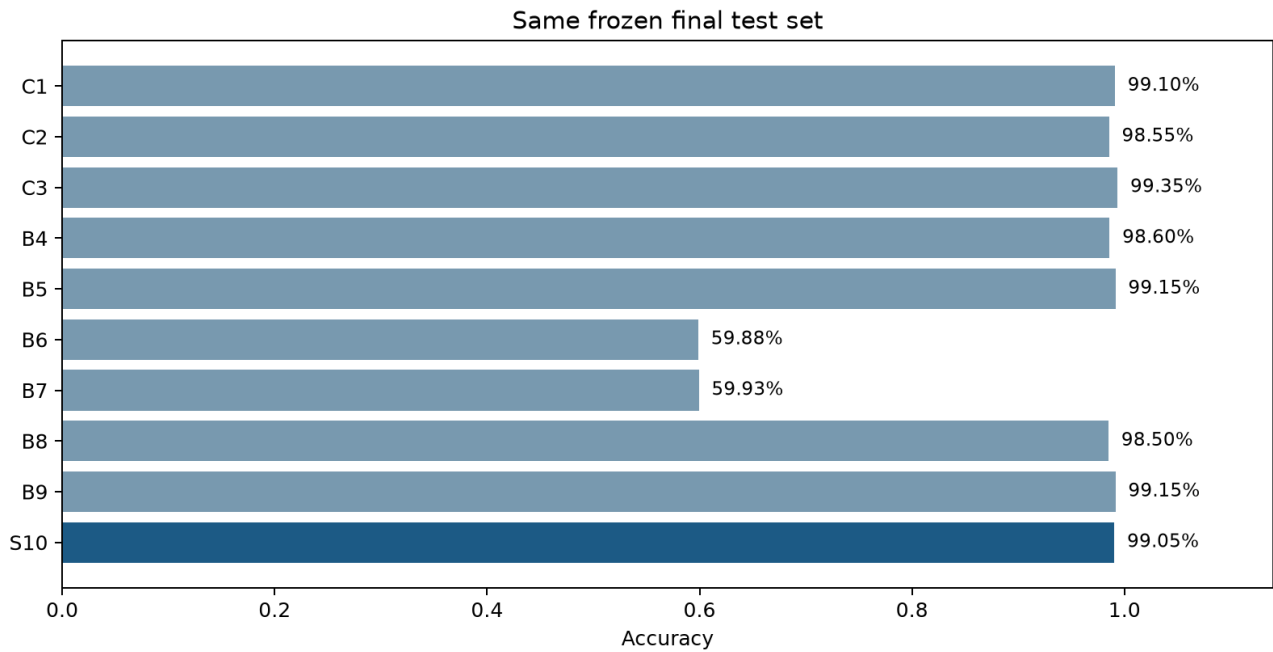
[L1] reports tenfold cross-validation of binary Korean severity classification: KTAS 3 versus KTAS 4-5. Recall and F1 are retained as published; they are not converted to our macro averages. BiLSTM and CNN are separate models, not a hybrid.

Our result uses three AI-assigned labels and a separate grouped holdout. These rows provide context, not evidence of superiority on a shared benchmark. The screenshot scores and its unidentified references [40]-[42] were not carried over because their sources could not be verified.

Reference

[L1] Seo et al. (2025). Artificial intelligence for severity triage based on conversations in an emergency department in Korea. Scientific Reports 15, 16870. Table 2. DOI: 10.1038/s41598-025-99874-0.

Final accuracy and macro F1



Final precision and ordinal errors

Ref.	Macro precision	Under-triage	Over-triage	Errors / records
C1	99.05%	0.45%	0.45%	18/1999
C2	98.46%	0.80%	0.65%	29/1999
C3	99.04%	0.65%	0.00%	13/1999
B4	98.07%	1.10%	0.30%	28/1999
B5	99.32%	0.20%	0.65%	17/1999
B6	64.10%	17.21%	22.91%	802/1999
B7	66.70%	21.41%	18.66%	801/1999
B8	98.03%	1.10%	0.40%	30/1999
B9	99.24%	0.30%	0.55%	17/1999
S10	99.01%	0.45%	0.50%	19/1999

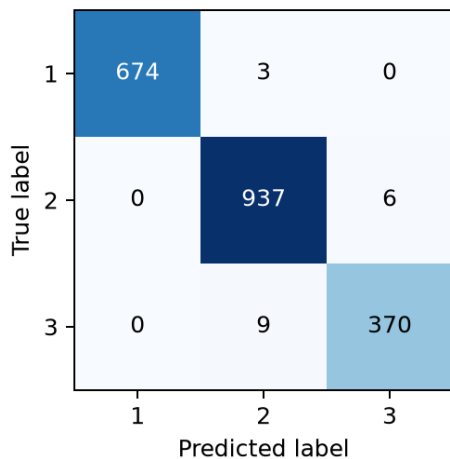
Under-triage predicts a numerically higher level (less urgency) than the supplied label. Over-triage predicts a lower level. These terms describe disagreement with the CSV labels, not independently reviewed patient harm.

QWK is quadratic weighted kappa; MAE is the average absolute difference between predicted and supplied label numbers. Accuracy alone can hide class-specific errors, so precision, recall, F1 and all confusion matrices are included.

Confusion matrices and class-level results

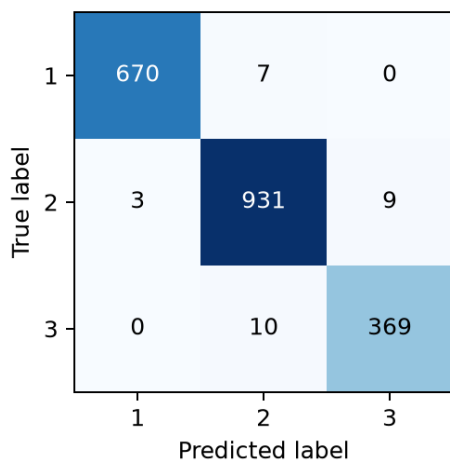
True labels are rows; predicted labels are columns. All matrices use the same final test set. Support is the number of records carrying each supplied label.

C1: Hist Gradient Boosting; SapBERT concepts + patient features / PCA-128 / derived features



Class	Precision	Recall	F1	N
1	100.00%	99.56%	99.78%	677
2	98.74%	99.36%	99.05%	943
3	98.40%	97.63%	98.01%	379
macro avg	99.05%	98.85%	98.95%	1999
weighted avg	99.10%	99.10%	99.10%	1999

C2: Logistic Regression; SapBERT concepts + patient features / full dimensions

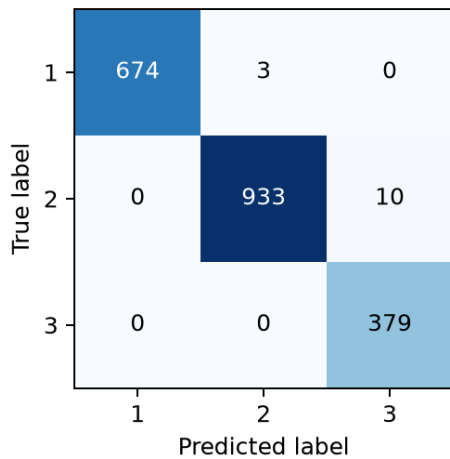


Class	Precision	Recall	F1	N
1	99.55%	98.97%	99.26%	677
2	98.21%	98.73%	98.47%	943
3	97.62%	97.36%	97.49%	379
macro avg	98.46%	98.35%	98.41%	1999
weighted avg	98.55%	98.55%	98.55%	1999

Confusion matrices and class-level results

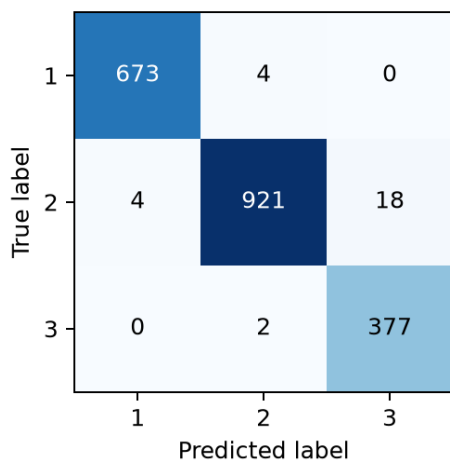
True labels are rows; predicted labels are columns. All matrices use the same final test set. Support is the number of records carrying each supplied label.

C3: Random Forest; SapBERT concepts + patient features / PCA-64



Class	Precision	Recall	F1	N
1	100.00%	99.56%	99.78%	677
2	99.68%	98.94%	99.31%	943
3	97.43%	100.00%	98.70%	379
macro avg	99.04%	99.50%	99.26%	1999
weighted avg	99.36%	99.35%	99.35%	1999

B4: Logistic Regression; Patient features

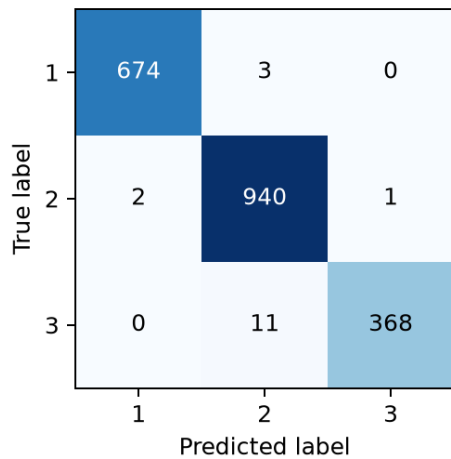


Class	Precision	Recall	F1	N
1	99.41%	99.41%	99.41%	677
2	99.35%	97.67%	98.50%	943
3	95.44%	99.47%	97.42%	379
macro avg	98.07%	98.85%	98.44%	1999
weighted avg	98.63%	98.60%	98.60%	1999

Confusion matrices and class-level results

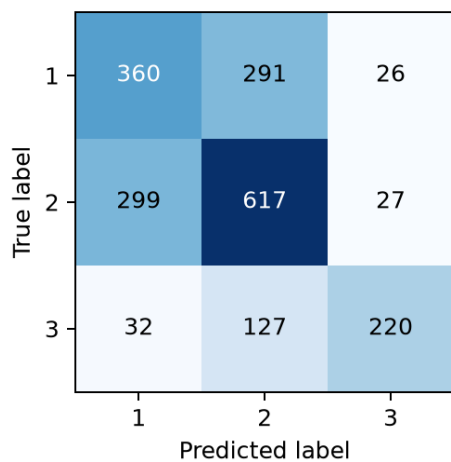
True labels are rows; predicted labels are columns. All matrices use the same final test set. Support is the number of records carrying each supplied label.

B5: Hist Gradient Boosting; Patient features



Class	Precision	Recall	F1	N
1	99.70%	99.56%	99.63%	677
2	98.53%	99.68%	99.10%	943
3	99.73%	97.10%	98.40%	379
macro avg	99.32%	98.78%	99.04%	1999
weighted avg	99.16%	99.15%	99.15%	1999

B6: Logistic Regression; SapBERT concepts / PCA-64

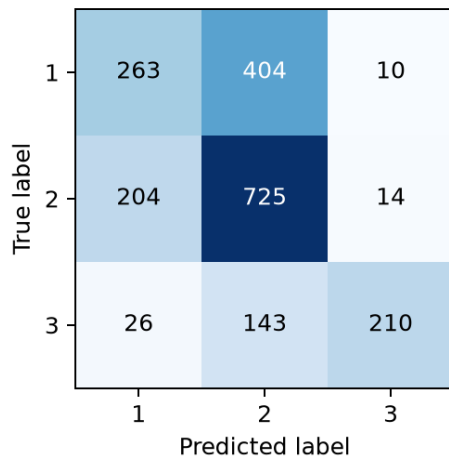


Class	Precision	Recall	F1	N
1	52.10%	53.18%	52.63%	677
2	59.61%	65.43%	62.39%	943
3	80.59%	58.05%	67.48%	379
macro avg	64.10%	58.88%	60.83%	1999
weighted avg	61.04%	59.88%	60.05%	1999

Confusion matrices and class-level results

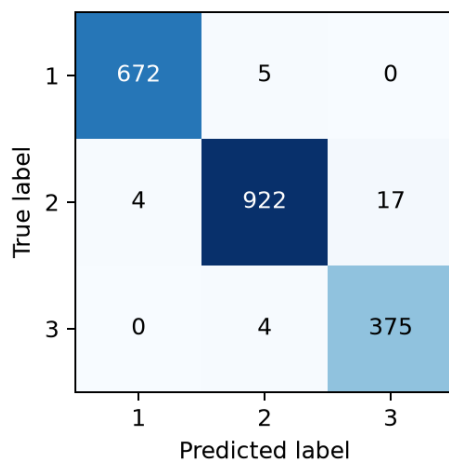
True labels are rows; predicted labels are columns. All matrices use the same final test set. Support is the number of records carrying each supplied label.

B7: Hist Gradient Boosting; SapBERT concepts / PCA-64



Class	Precision	Recall	F1	N
1	53.35%	38.85%	44.96%	677
2	57.00%	76.88%	65.46%	943
3	89.74%	55.41%	68.52%	379
macro avg	66.70%	57.05%	59.65%	1999
weighted avg	61.97%	59.93%	59.10%	1999

B8: Logistic Regression; SapBERT concepts + patient features / PCA-64

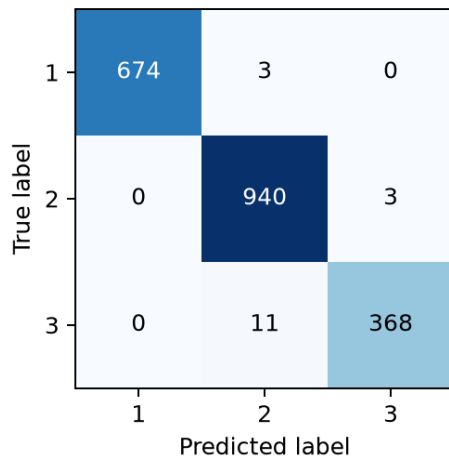


Class	Precision	Recall	F1	N
1	99.41%	99.26%	99.33%	677
2	99.03%	97.77%	98.40%	943
3	95.66%	98.94%	97.28%	379
macro avg	98.03%	98.66%	98.34%	1999
weighted avg	98.52%	98.50%	98.50%	1999

Confusion matrices and class-level results

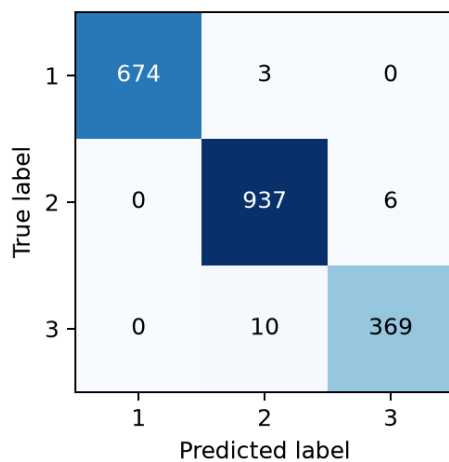
True labels are rows; predicted labels are columns. All matrices use the same final test set. Support is the number of records carrying each supplied label.

B9: Hist Gradient Boosting; SapBERT concepts + patient features / PCA-64



Class	Precision	Recall	F1	N
1	100.00%	99.56%	99.78%	677
2	98.53%	99.68%	99.10%	943
3	99.19%	97.10%	98.13%	379
macro avg	99.24%	98.78%	99.01%	1999
weighted avg	99.15%	99.15%	99.15%	1999

S10: Selected system



Class	Precision	Recall	F1	N
1	100.00%	99.56%	99.78%	677
2	98.63%	99.36%	99.00%	943
3	98.40%	97.36%	97.88%	379
macro avg	99.01%	98.76%	98.88%	1999
weighted avg	99.05%	99.05%	99.05%	1999

Sensitivity checks and dataset patterns

Removed features	CV accuracy	CV macro F1
ECG_Status	97.48%	97.38%
ECG_Status, Mode_of_Arrival	97.43%	97.37%
Gender	98.95%	98.78%

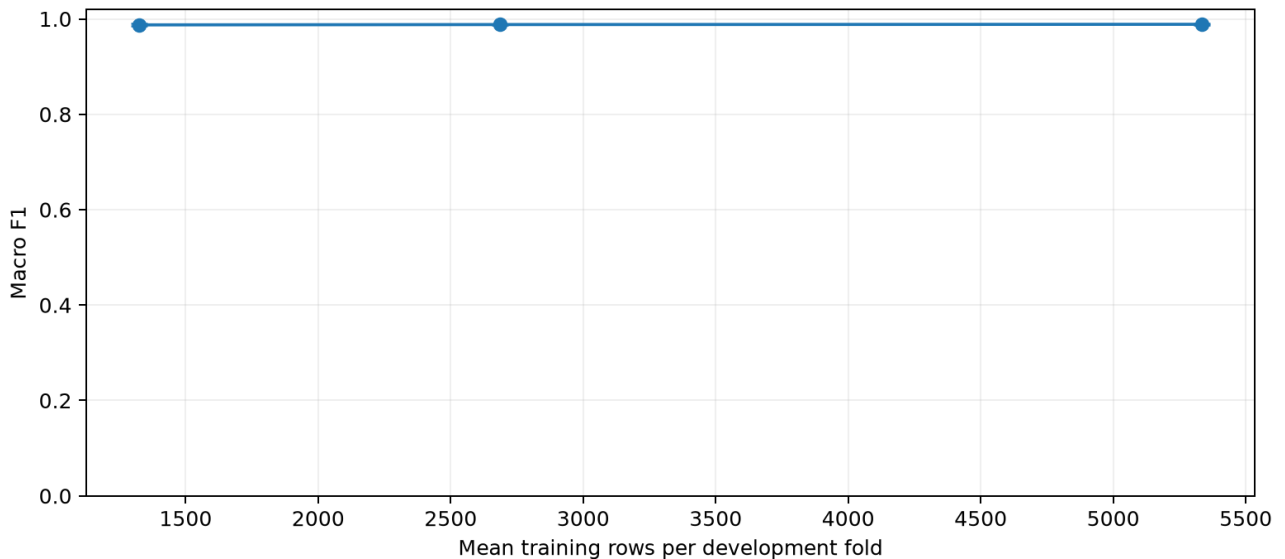
These are three-fold grouped development comparisons using the same boosting settings. The presence of a predictive input is not automatically leakage: ECG and vital signs are legitimate prediction-time features. However, simple label-specific patterns suggest the dataset may be easier than new clinical cases.

ECG category	Label 1	Label 2	Label 3
Abnormal	451	628	380
Anterior MI	492	0	0
Arrhythmia	0	601	0
Atrial fibrillation	0	634	0
Inferior MI	458	0	0
Normal	0	0	362
ST depression	0	609	0
ST elevation	1311	0	0
Sinus tachycardia	0	660	407
T wave inversion	0	641	367

This table uses development records only. Several ECG categories occur under just one label. Numeric feature ranges also differ strongly by label; they are retained in `development_feature_ranges.csv`. The AI-assigned label source is user-confirmed. Independent clinician-labelled data is needed before interpreting strong internal scores as general clinical performance.

Learning curve and label-shuffle control

Training-group fraction	Mean rows	Accuracy	Macro F1	F1 std.
25.00%	1326	98.90%	98.74%	0.0027
50.00%	2685	98.98%	98.81%	0.0008
100.00%	5334	99.03%	98.87%	0.0011



Training-label shuffle control

With only training labels randomly permuted, mean development accuracy was 42.42% and macro F1 was 32.41%. Validation labels were not changed. This is a diagnostic control, not a candidate model or a complete proof against every possible form of leakage.

Protocol assertions also verified group separation in every split, no exact normalized text overlap in the final split, training-only PCA means, handling of unseen categories, and invariance of model features when target labels are changed.

5-fold grouped CV: refined models

Candidates 1-22 of 22. Scores are means across development folds; SD is the population standard deviation of fold macro F1. These scores guided selection and should not be treated as independent final-test estimates.

Trial	Model / inputs	PCA	Accuracy	Macro F1	F1 SD
ebf1fb91	hgb / fused Sap	128	99.28%	99.17%	0.0010
0f3df5b6	hgb / fused Sap	64	99.25%	99.14%	0.0012
a55cdf81	hgb / fused MPNet	64	99.24%	99.12%	0.0022
74be9299	hgb / fused MiniLM	32	99.24%	99.12%	0.0008
e905b91d	hgb / fused MiniLM	64	99.23%	99.09%	0.0014
8e1de632	hgb / fused MiniLM	128	99.21%	99.08%	0.0019
f8d183a4	hgb / fused Sap	32	99.19%	99.07%	0.0013
2a938b56	hgb / fused MiniLM	32	99.20%	99.06%	0.0020
37444b14	rf / fused Sap	64	99.09%	98.94%	0.0020
38acfd8	rf / structured	-	99.04%	98.88%	0.0016
96313af0	hgb / structured	-	98.87%	98.70%	0.0023
bbd4c7fd	logreg / fused Sap	full	98.46%	98.37%	0.0035
a3b46c2b	logreg / fused Sap	full	98.38%	98.29%	0.0034
6b4606af	logreg / fused Sap	full	98.34%	98.25%	0.0041
254c1f79	logreg / fused Sap	full	98.31%	98.22%	0.0040
245e9d6c	logreg / fused Sap	full	98.31%	98.22%	0.0040
92b5a9e2	logreg / fused Sap	full	98.26%	98.13%	0.0035
99ada6ae	logreg / fused Sap	full	98.16%	98.03%	0.0034
4cf94fa0	logreg / structured	-	98.14%	98.00%	0.0045
a1ff41d7	hgb / text MiniLM	64	64.05%	63.04%	0.0182
e5e4ef77	logreg / text MiniLM	full	61.48%	61.48%	0.0121
a7d542d8	rf / text MiniLM	64	61.15%	60.07%	0.0206

Trial IDs map to saved configuration JSON files, including class weights, regularization, tree settings, derived features and text weights. Full-precision per-fold accuracy, precision, recall, F1, QWK, MAE and directional-error values are available in cv3/ and cv5/.

3-fold grouped CV: screening candidates

Candidates 1-23 of 136. Scores are means across development folds; SD is the population standard deviation of fold macro F1. These scores guided selection and should not be treated as independent final-test estimates.

Trial	Model / inputs	PCA	Accuracy	Macro F1	F1 SD
2a938b56	hgb / fused MiniLM	32	99.33%	99.23%	0.0009
74be9299	hgb / fused MiniLM	32	99.30%	99.20%	0.0008
e905b91d	hgb / fused MiniLM	64	99.26%	99.17%	0.0005
f8d183a4	hgb / fused Sap	32	99.28%	99.16%	0.0008
0f3df5b6	hgb / fused Sap	64	99.26%	99.16%	0.0014
8e1de632	hgb / fused MiniLM	128	99.25%	99.14%	0.0006
ebf1fb91	hgb / fused Sap	128	99.25%	99.13%	0.0006
a55cdf81	hgb / fused MPNet	64	99.24%	99.12%	0.0009
8515b026	hgb / fused MiniLM	128	99.24%	99.12%	0.0003
f017b881	hgb / fused Sap	32	99.23%	99.12%	0.0003
21335653	hgb / fused MiniLM	64	99.23%	99.11%	0.0001
5a74e870	hgb / fused Sap	128	99.23%	99.11%	0.0006
83cb14ae	hgb / fused MPNet	64	99.20%	99.09%	0.0018
9b4ffa16	hgb / fused Sap	64	99.21%	99.09%	0.0011
39527dcf	hgb / fused MPNet	128	99.20%	99.07%	0.0013
54fa3c8d	hgb / fused MPNet	128	99.18%	99.05%	0.0020
65244325	hgb / fused Sap	64	99.18%	99.05%	0.0013
1e3a0be2	hgb / fused MPNet	32	99.15%	99.00%	0.0006
b6f4363e	hgb / fused MPNet	32	99.14%	98.99%	0.0011
38acfd8	rf / structured	-	99.11%	98.97%	0.0018
0667184a	rf / fused MPNet	64	99.09%	98.94%	0.0018
5f798c87	rf / fused MPNet	128	99.09%	98.94%	0.0018
a452511e	rf / fused MiniLM	64	99.09%	98.94%	0.0018

Trial IDs map to saved configuration JSON files, including class weights, regularization, tree settings, derived features and text weights. Full-precision per-fold accuracy, precision, recall, F1, QWK, MAE and directional-error values are available in cv3/ and cv5/.

3-fold grouped CV: screening candidates

Candidates 24-46 of 136. Scores are means across development folds; SD is the population standard deviation of fold macro F1. These scores guided selection and should not be treated as independent final-test estimates.

Trial	Model / inputs	PCA	Accuracy	Macro F1	F1 SD
a6967740	rf / fused Sap	128	99.09%	98.94%	0.0018
37444b14	rf / fused Sap	64	99.09%	98.94%	0.0018
7646ab1b	rf / fused MiniLM	128	99.09%	98.94%	0.0018
6ee14e41	rf / structured	-	99.03%	98.91%	0.0013
96313af0	hgb / structured	-	99.04%	98.88%	0.0012
7efadcb4	hgb / structured	-	99.03%	98.87%	0.0006
67aed8b0	hgb / structured	-	99.03%	98.87%	0.0009
96bf1615	hgb / structured	-	99.01%	98.86%	0.0013
5dbe30c4	hgb / structured	-	99.01%	98.86%	0.0011
c71ab44d	hgb / structured	-	99.01%	98.85%	0.0005
87236b7b	hgb / structured	-	99.00%	98.85%	0.0009
08da2b40	hgb / structured	-	98.99%	98.82%	0.0006
6d070b39	hgb / structured	-	98.95%	98.79%	0.0008
0ea992ea	hgb / structured	-	98.95%	98.78%	0.0009
0f79df3e	hgb / structured	-	98.94%	98.77%	0.0009
5e44160f	hgb / structured	-	98.95%	98.77%	0.0017
6c11afaf	rf / structured	-	98.96%	98.77%	0.0023
86b91358	hgb / structured	-	98.94%	98.75%	0.0007
c8129735	rf / structured	-	98.91%	98.73%	0.0006
494602c5	rf / structured	-	98.83%	98.60%	0.0020
6ea35e5b	rf / structured	-	98.79%	98.59%	0.0004
245e9d6c	logreg / fused Sap	full	98.34%	98.24%	0.0026
178519df	logreg / fused Sap	32	98.33%	98.24%	0.0029

Trial IDs map to saved configuration JSON files, including class weights, regularization, tree settings, derived features and text weights. Full-precision per-fold accuracy, precision, recall, F1, QWK, MAE and directional-error values are available in cv3/ and cv5/.

3-fold grouped CV: screening candidates

Candidates 47-69 of 136. Scores are means across development folds; SD is the population standard deviation of fold macro F1. These scores guided selection and should not be treated as independent final-test estimates.

Trial	Model / inputs	PCA	Accuracy	Macro F1	F1 SD
e16f8744	logreg / fused Sap	64	98.33%	98.23%	0.0029
2da0a205	logreg / fused Sap	32	98.31%	98.22%	0.0019
a91e764c	logreg / fused Sap	128	98.31%	98.21%	0.0028
5c66f0e8	logreg / fused MPNet	full	98.30%	98.20%	0.0034
2d1828be	logreg / fused MPNet	128	98.28%	98.17%	0.0032
82d798df	logreg / fused MPNet	64	98.26%	98.15%	0.0031
823ceb59	logreg / fused Sap	full	98.26%	98.15%	0.0014
2554e14b	logreg / fused Sap	64	98.23%	98.11%	0.0013
d2f91a10	logreg / fused MPNet	128	98.23%	98.11%	0.0035
3d51a786	logreg / fused Sap	128	98.21%	98.10%	0.0012
98b93fcf	logreg / fused MPNet	64	98.21%	98.09%	0.0033
c2d5dbbf	logreg / fused MPNet	full	98.21%	98.09%	0.0036
0148a626	logreg / fused MiniLM	full	98.18%	98.07%	0.0027
4cf94fa0	logreg / structured	-	98.19%	98.05%	0.0031
bb758424	logreg / fused MiniLM	128	98.15%	98.04%	0.0023
d2b50391	logreg / fused MiniLM	32	98.16%	98.03%	0.0036
99d8a9c2	logreg / fused MiniLM	32	98.13%	98.01%	0.0023
0131a3ae	logreg / structured	-	98.15%	98.01%	0.0025
09d7b6e4	logreg / fused MiniLM	full	98.11%	97.98%	0.0029
bd5cf19f	logreg / fused MiniLM	64	98.11%	97.98%	0.0031
48b585ac	logreg / fused MiniLM	128	98.10%	97.97%	0.0030
32f426eb	logreg / structured	-	98.10%	97.97%	0.0030
a3b3ccd2	logreg / fused MiniLM	64	98.08%	97.97%	0.0018

Trial IDs map to saved configuration JSON files, including class weights, regularization, tree settings, derived features and text weights. Full-precision per-fold accuracy, precision, recall, F1, QWK, MAE and directional-error values are available in cv3/ and cv5/.

3-fold grouped CV: screening candidates

Candidates 70-92 of 136. Scores are means across development folds; SD is the population standard deviation of fold macro F1. These scores guided selection and should not be treated as independent final-test estimates.

Trial	Model / inputs	PCA	Accuracy	Macro F1	F1 SD
8719c885	logreg / structured	-	98.11%	97.96%	0.0053
413150e2	logreg / structured	-	98.11%	97.95%	0.0054
84e1c3cf	logreg / structured	-	98.08%	97.94%	0.0028
a06b6931	logreg / structured	-	98.09%	97.92%	0.0057
c92f703a	logreg / fused MPNet	32	98.06%	97.92%	0.0028
bb1db875	logreg / structured	-	98.08%	97.91%	0.0056
a8be55dd	logreg / fused Sap	32	98.01%	97.89%	0.0037
ec1ae00f	logreg / fused MPNet	32	98.04%	97.89%	0.0026
ffdcce7	logreg / fused Sap	full	98.01%	97.88%	0.0038
d8eae668	logreg / structured	-	98.05%	97.88%	0.0046
09fc408c	logreg / fused Sap	128	97.99%	97.86%	0.0035
213eb1f2	logreg / fused Sap	64	97.99%	97.86%	0.0035
49f3e843	logreg / structured	-	97.99%	97.82%	0.0045
dbb615a8	logreg / fused MiniLM	full	97.95%	97.82%	0.0029
a70fb306	logreg / fused MiniLM	32	97.91%	97.77%	0.0029
737dce83	logreg / fused MiniLM	64	97.91%	97.77%	0.0029
93aaf7b5	logreg / fused MiniLM	128	97.90%	97.76%	0.0027
c67dfafd	logreg / fused MPNet	64	97.86%	97.71%	0.0029
bd500b84	logreg / fused MPNet	128	97.85%	97.70%	0.0031
7b9f6c0e	logreg / fused MPNet	full	97.84%	97.69%	0.0030
f24fd817	logreg / fused MPNet	32	97.80%	97.65%	0.0032
f3809a6b	logreg / structured	-	97.79%	97.63%	0.0020
ec8e4f23	logreg / structured	-	97.78%	97.61%	0.0037

Trial IDs map to saved configuration JSON files, including class weights, regularization, tree settings, derived features and text weights. Full-precision per-fold accuracy, precision, recall, F1, QWK, MAE and directional-error values are available in cv3/ and cv5/.

3-fold grouped CV: screening candidates

Candidates 93-115 of 136. Scores are means across development folds; SD is the population standard deviation of fold macro F1. These scores guided selection and should not be treated as independent final-test estimates.

Trial	Model / inputs	PCA	Accuracy	Macro F1	F1 SD
7c5b0b2f	hgb / structured	-	97.48%	97.38%	0.0016
2866a161	hgb / structured	-	97.43%	97.37%	0.0010
a1ff41d7	hgb / text MiniLM	64	63.54%	62.31%	0.0091
e5e4ef77	logreg / text MiniLM	full	61.72%	61.70%	0.0052
f2bab23e	logreg / text MiniLM	full	61.10%	61.46%	0.0041
34a1a161	logreg / text MiniLM	128	60.98%	61.36%	0.0030
40e4af31	logreg / text MiniLM	128	60.87%	60.97%	0.0017
3222777c	logreg / text MiniLM	64	60.10%	60.33%	0.0005
a7d542d8	rf / text MiniLM	64	60.88%	59.94%	0.0088
b20bb592	logreg / text MiniLM	64	59.44%	59.92%	0.0043
328b585e	hgb / text Sap	64	60.45%	59.81%	0.0041
06344ef1	logreg / text Sap	128	59.27%	59.63%	0.0048
7b7a15f2	logreg / text Sap	64	59.09%	59.62%	0.0048
245c317a	logreg / text Sap	full	59.26%	59.54%	0.0059
4e0ee917	logreg / text Sap	64	58.79%	59.23%	0.0040
bf6d38a1	logreg / text Sap	128	58.58%	59.18%	0.0046
ed11eb2b	logreg / text Sap	full	58.56%	59.16%	0.0035
daa2fb39	logreg / text Sap	64	58.37%	58.99%	0.0047
24773bc4	logreg / text Sap	32	58.18%	58.60%	0.0021
15a5c464	logreg / text Sap	32	57.84%	58.49%	0.0054
459be584	rf / text Sap	64	58.80%	58.29%	0.0048
c4df0c5d	logreg / text Sap	32	57.72%	58.06%	0.0044
a9ebda5d	logreg / text MPNet	full	58.48%	58.05%	0.0112

Trial IDs map to saved configuration JSON files, including class weights, regularization, tree settings, derived features and text weights. Full-precision per-fold accuracy, precision, recall, F1, QWK, MAE and directional-error values are available in cv3/ and cv5/.

3-fold grouped CV: screening candidates

Candidates 116-136 of 136. Scores are means across development folds; SD is the population standard deviation of fold macro F1. These scores guided selection and should not be treated as independent final-test estimates.

Trial	Model / inputs	PCA	Accuracy	Macro F1	F1 SD
62977ce3	logreg / text Sap	128	57.53%	57.96%	0.0021
bb82cfad	hgb / text MPNet	64	58.81%	57.76%	0.0076
b1f22bf1	logreg / text Sap	full	57.44%	57.62%	0.0104
6e95bd80	logreg / text MPNet	full	57.66%	57.56%	0.0098
ef72fa6c	logreg / text MPNet	128	57.68%	57.40%	0.0074
c39e7954	logreg / text MPNet	128	57.38%	57.24%	0.0088
0e341427	logreg / text MiniLM	full	56.49%	56.86%	0.0082
5685a791	logreg / text MiniLM	128	56.44%	56.82%	0.0087
7d6c7832	logreg / text MPNet	64	56.86%	56.58%	0.0104
143cdb6e	logreg / text MiniLM	64	55.84%	56.30%	0.0102
250580ca	logreg / text MPNet	64	56.21%	56.10%	0.0090
1a650444	rf / text MPNet	64	56.98%	55.88%	0.0010
8fc6d6d5	logreg / text MiniLM	32	54.91%	55.77%	0.0037
9e30ac74	logreg / text MiniLM	32	54.87%	55.68%	0.0047
5b412211	logreg / text MiniLM	32	53.07%	53.64%	0.0074
c84876f4	logreg / text MPNet	128	51.89%	52.03%	0.0056
607de0d4	logreg / text MPNet	full	51.86%	51.98%	0.0061
6887fe93	logreg / text MPNet	64	51.31%	51.42%	0.0049
6cc90ef2	logreg / text MPNet	32	47.14%	47.09%	0.0081
07c76c97	logreg / text MPNet	32	46.86%	46.82%	0.0073
e041fdb9	logreg / text MPNet	32	45.16%	45.09%	0.0098

Trial IDs map to saved configuration JSON files, including class weights, regularization, tree settings, derived features and text weights. Full-precision per-fold accuracy, precision, recall, F1, QWK, MAE and directional-error values are available in cv3/ and cv5/.

Selected settings and reproducibility

Trial ebf1fb91109f52; ensemble weight 1.000

Hist Gradient Boosting; SapBERT concepts + patient features / PCA-128 / derived features

Classifier parameters: {"max_iter": 250, "max_leaf_nodes": 15, "l2_regularization": 1, "learning_rate": 0.1, "balance": true}

Feature parameters: {"view": "fused", "encoder": "sapbert_concept", "pca": 128, "derived": true}

Reproduction files

prepare_data.py; encode_text.py; research_engine.py; validate_protocol.py; sensitivity.py; diagnostics.py; run_study.py; build_report.py. The uploaded Python file is preserved as triage_classifier_uploaded.py. Data checksums, split assignments, encoder revisions, search plans, per-fold predictions, frozen selection, final predictions, fitted preprocessors and classifiers are retained alongside this report.

The saved models are research artifacts. Their feature objects refer to row-aligned embedding arrays in the experiment directory; they are not a replacement for the application's live translation/inference pipeline. Supporting files must be kept with the models to reproduce predictions on these records.

Boundaries of the study

The study covers only labels 1-3 and one uploaded dataset. It does not demonstrate level-4 performance, live Roman Urdu translation quality, calibration, temporal generalization, clinical usefulness, or literature superiority. No published-comparator scores are invented. Strong internal scores require independent data and verified label provenance before broader claims.

Reported gains must be separated into dataset differences and within-dataset model differences. All final comparisons in this report use identical test rows; earlier reports used different records, class counts and split protocols.

Appendix A: original SBERT comparison

Historical report reproduced unchanged. Its dataset, target classes, split and scores belong to the earlier experiment. Do not interpret its values as a controlled comparison with the new 10,000-record study.

SBERT Classifier Comparison

Sections 3.4 and 3.5 | Research experiment results

Encoder: sentence-transformers/all-mpnet-base-v2. Frozen, normalized 768-dimensional embeddings of existing English translations. PCA reduces the text representation to 64 dimensions.

Dataset: 2,252 synthetic cardiac records. Stratified split: 1,801 training and 451 test rows; seed 42. Every comparison uses the same test rows. PCA and patient-feature preprocessing are fitted on training rows only.

Text-only comparison

Text size	Classifier	Accuracy	Precision*	F1*
768-D	Logistic Regression	58.09%	53.49%	54.85%
768-D	Hist Gradient Boosting	57.43%	53.29%	47.98%
768-D	Random Forest	55.65%	55.53%	51.08%
PCA-64	Logistic Regression	55.65%	50.68%	51.78%
PCA-64	Hist Gradient Boosting	56.98%	51.39%	47.34%
PCA-64	Random Forest	54.32%	47.86%	46.53%

Text plus patient features

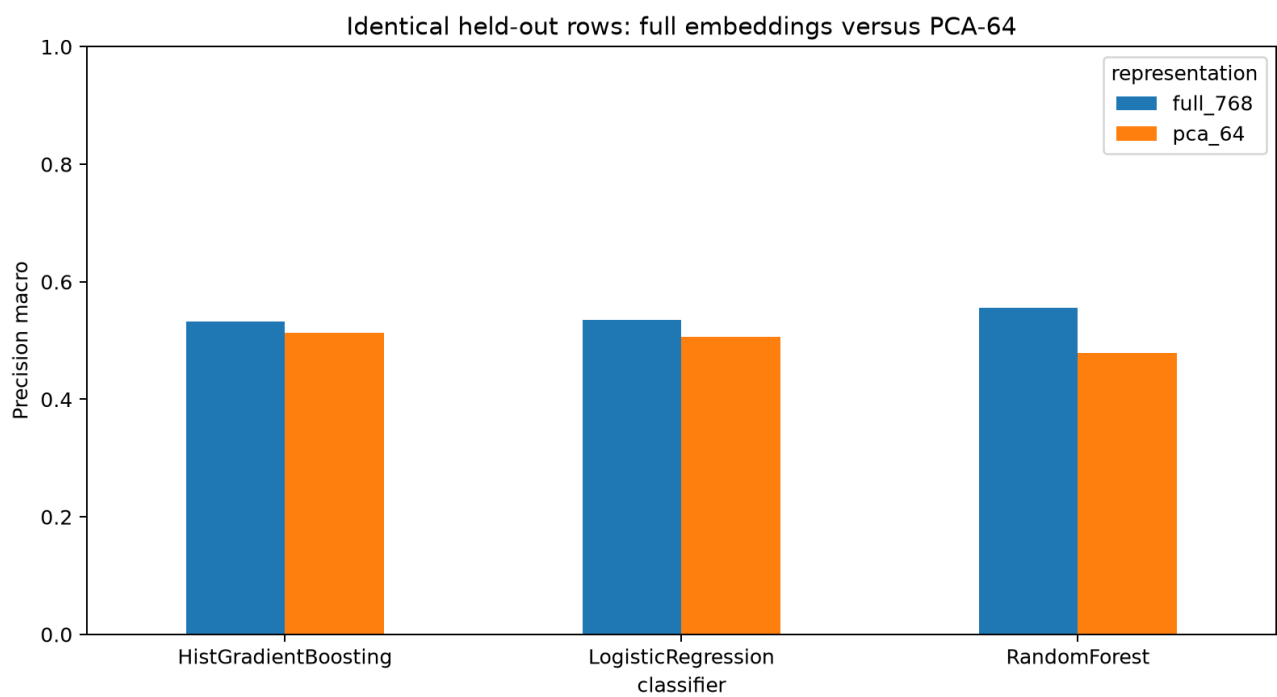
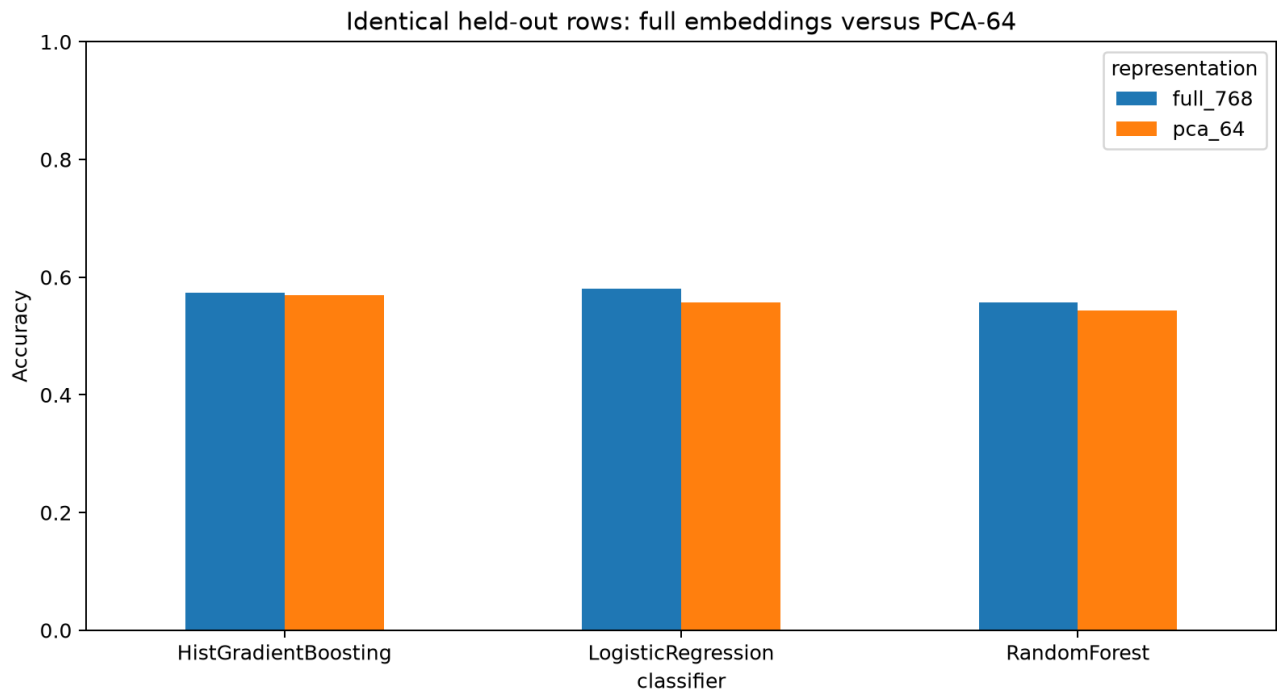
Text size	Classifier	Accuracy	Precision*	F1*
768-D	Logistic Regression	82.48%	77.03%	78.68%
768-D	Hist Gradient Boosting	82.93%	82.08%	76.97%
768-D	Random Forest	72.73%	70.12%	64.37%
PCA-64	Logistic Regression	82.26%	76.58%	78.28%
PCA-64	Hist Gradient Boosting	80.71%	81.54%	77.27%
PCA-64	Random Forest	80.71%	79.13%	73.06%

*Precision and F1 are macro-averaged. Fusion adds patient features to the stated text dimension. PCA retains 89.71% of training embedding variance.

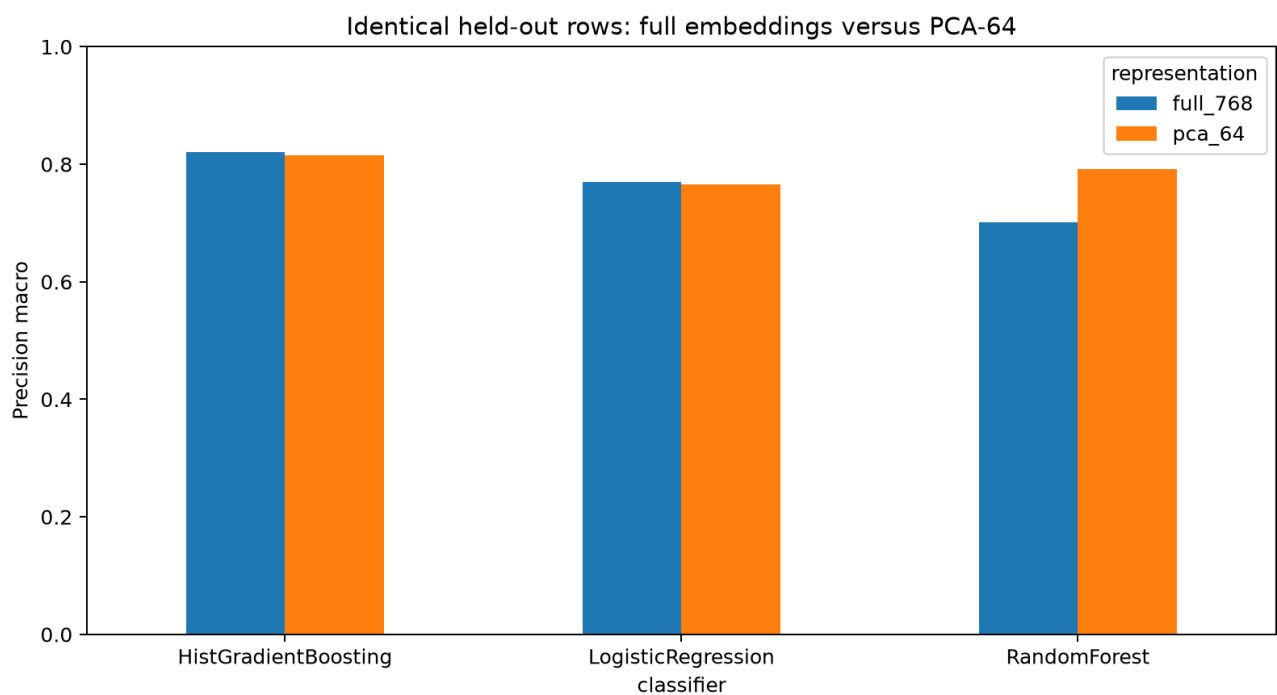
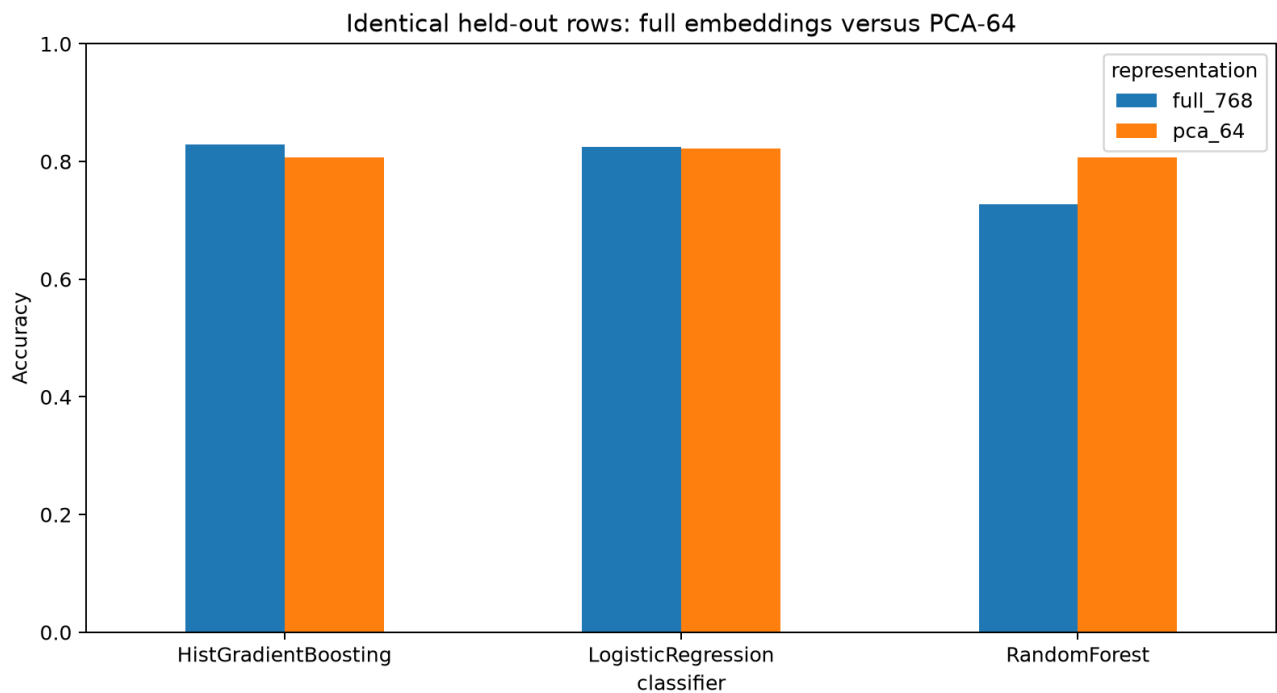
Interpretation and Section 3.6

These are single-split synthetic-data results, not clinical validation or an evaluation of live translation. Repeated complaint phrases may cross the random row split. Section 3.6 remains pending: the article has only a heading and no published comparator list. No literature scores have been invented.

Text-only performance

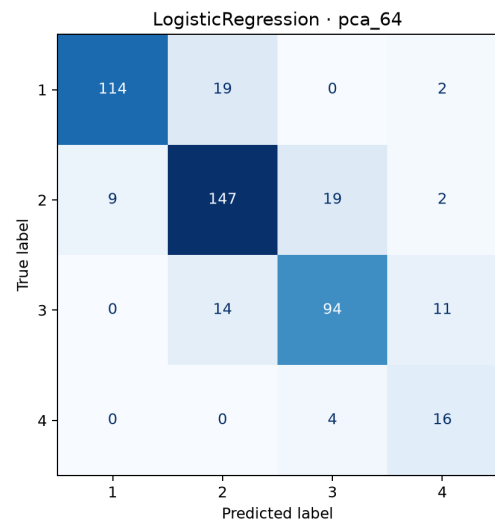
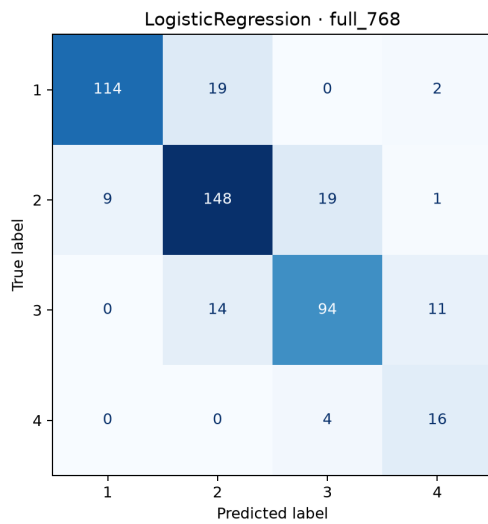
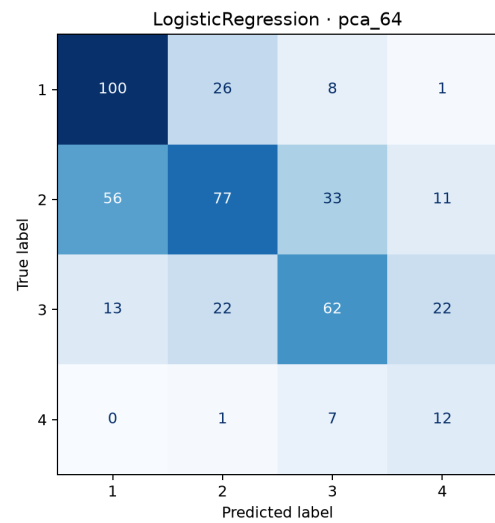
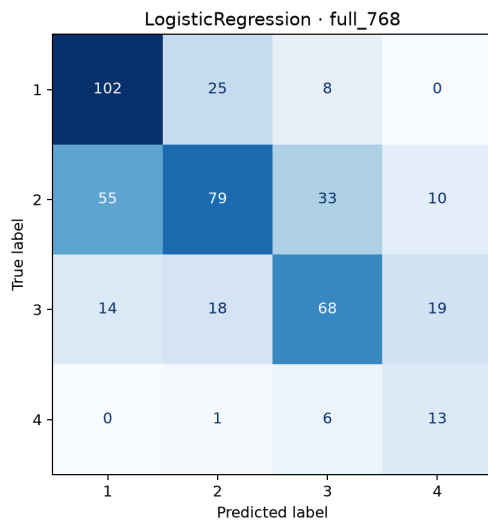


Performance with patient features



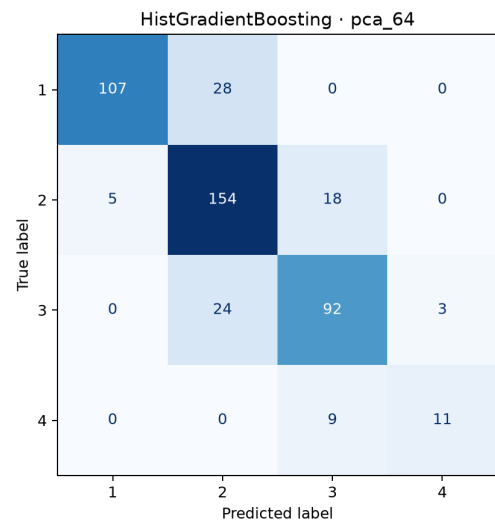
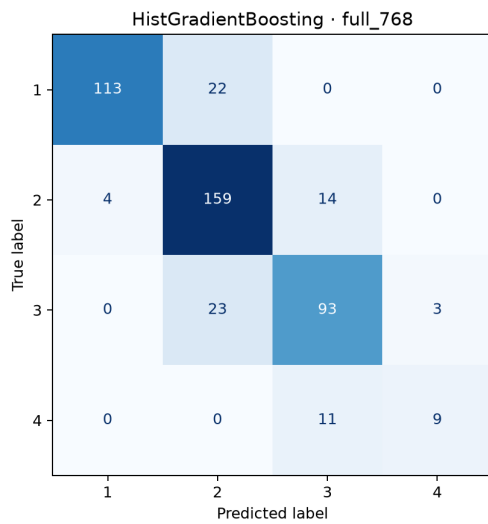
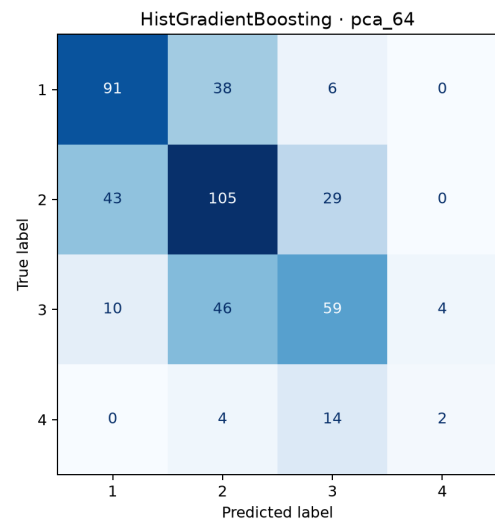
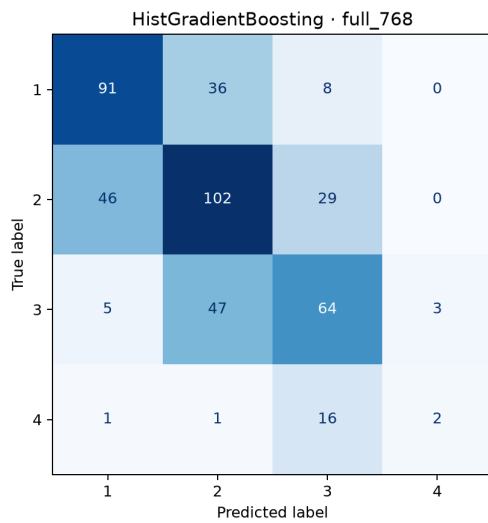
Logistic Regression: confusion matrices

Rows show true triage levels; columns show predictions. Each matrix contains the same 451 test records. Top row: text only. Bottom row: text plus patient features. Left: original 768-D; right: PCA-64.



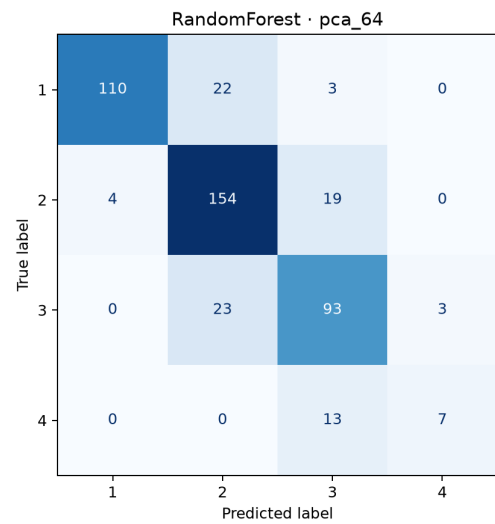
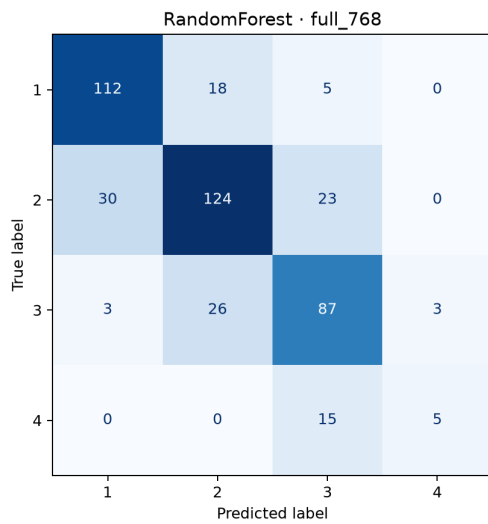
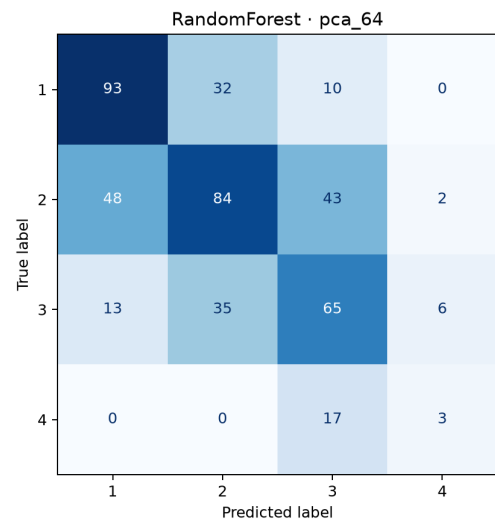
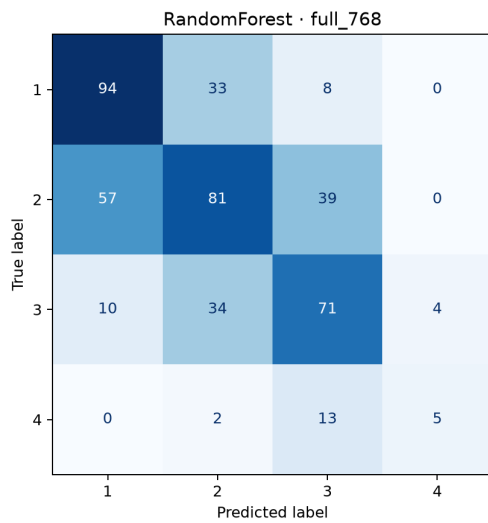
Hist Gradient Boosting: confusion matrices

Rows show true triage levels; columns show predictions. Each matrix contains the same 451 test records. Top row: text only. Bottom row: text plus patient features. Left: original 768-D; right: PCA-64.



Random Forest: confusion matrices

Rows show true triage levels; columns show predictions. Each matrix contains the same 451 test records. Top row: text only. Bottom row: text plus patient features. Left: original 768-D; right: PCA-64.



Appendix B: expanded SapBERT comparison

Historical report reproduced unchanged. Its dataset, target classes, split and scores belong to the earlier experiment. Do not interpret its values as a controlled comparison with the new 10,000-record study.

SapBERT Classifier Comparison

Sections 3.4 and 3.5 | Complete research experiment comparison

Based on the latest uploaded triage_classifier.py. Its SapBERT encoder, data preparation, patient-feature preprocessing, grouped split and Logistic Regression / Hist Gradient Boosting settings are retained. Original 768-D tests and Random Forest are explicitly added comparisons; the uploaded file remains unchanged.

Dataset: 821 synthetic cardiac records with existing generated clinical concepts. The first recorded concept for each complaint was matched to cardiac_english_2252.csv. Split: 609 training and 212 test records; no exact concept string occurs in both sets. All comparisons use these same test records.

Encoder: cambridgeltl/SapBERT-from-PubMedBERT-fulltext. CLS pooling, L2 normalization and 64-token truncation produce 768-D embeddings. Training-only PCA reduces these to 64 dimensions and retains 90.7% of variance. Fusion adds 24 patient features (792 or 88 total features).

Text-only comparison

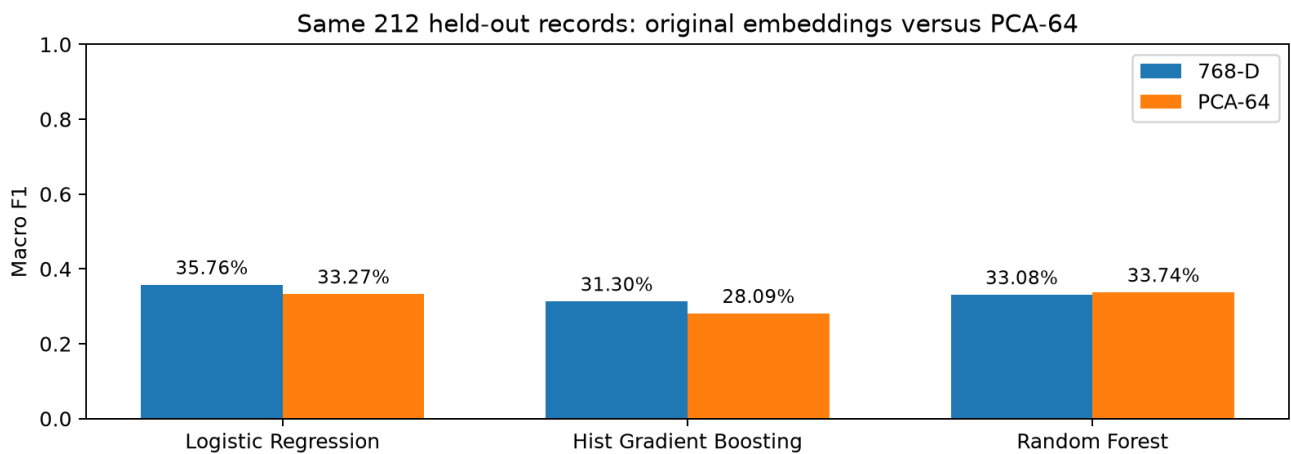
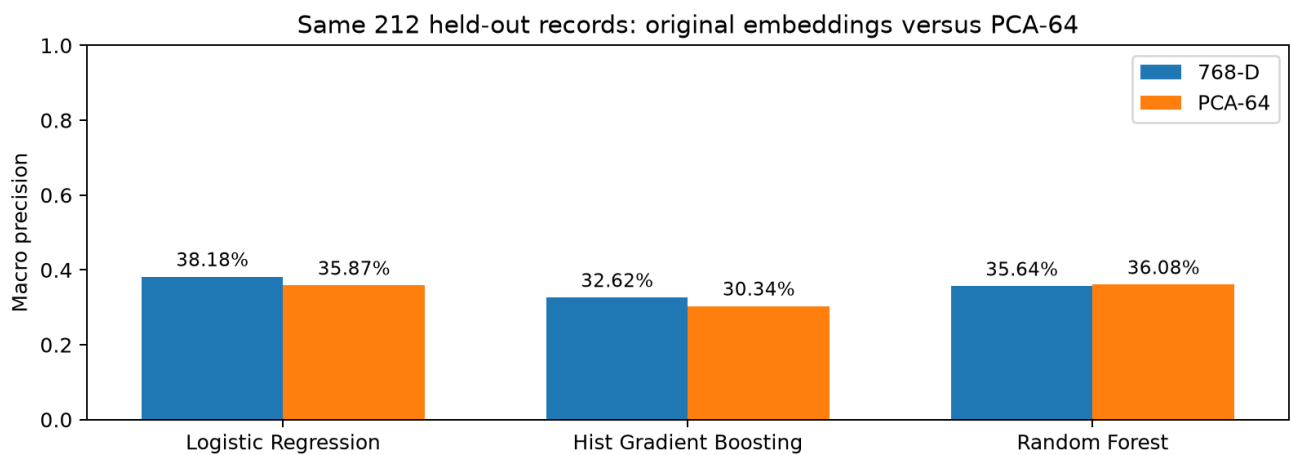
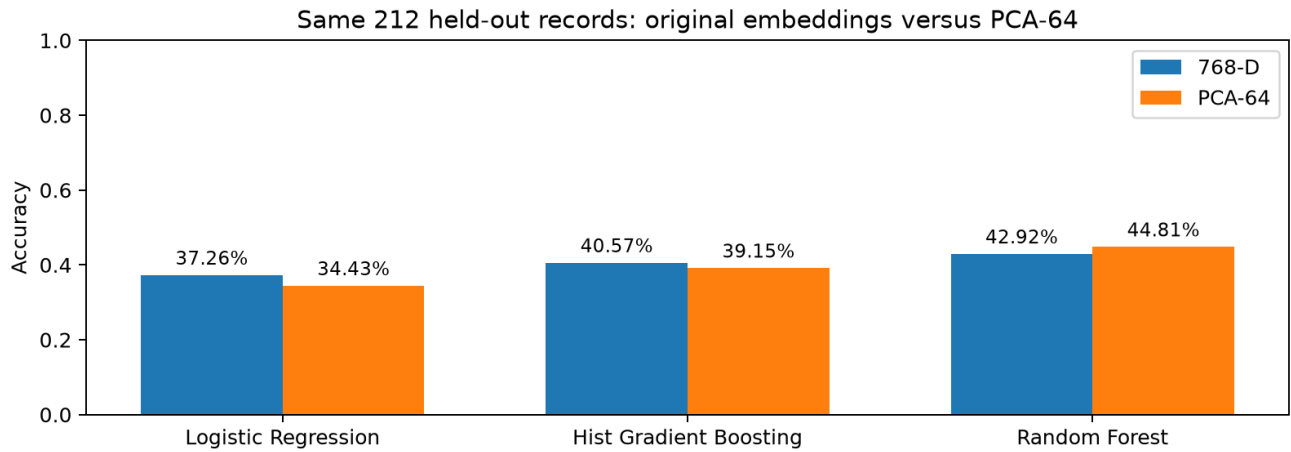
Text size	Classifier	Accuracy	Precision*	F1*
768-D	Logistic Regression	37.26%	38.18%	35.76%
768-D	Hist Gradient Boosting	40.57%	32.62%	31.30%
768-D	Random Forest	42.92%	35.64%	33.08%
PCA-64	Logistic Regression	34.43%	35.87%	33.27%
PCA-64	Hist Gradient Boosting	39.15%	30.34%	28.09%
PCA-64	Random Forest	44.81%	36.08%	33.74%

Text plus patient features

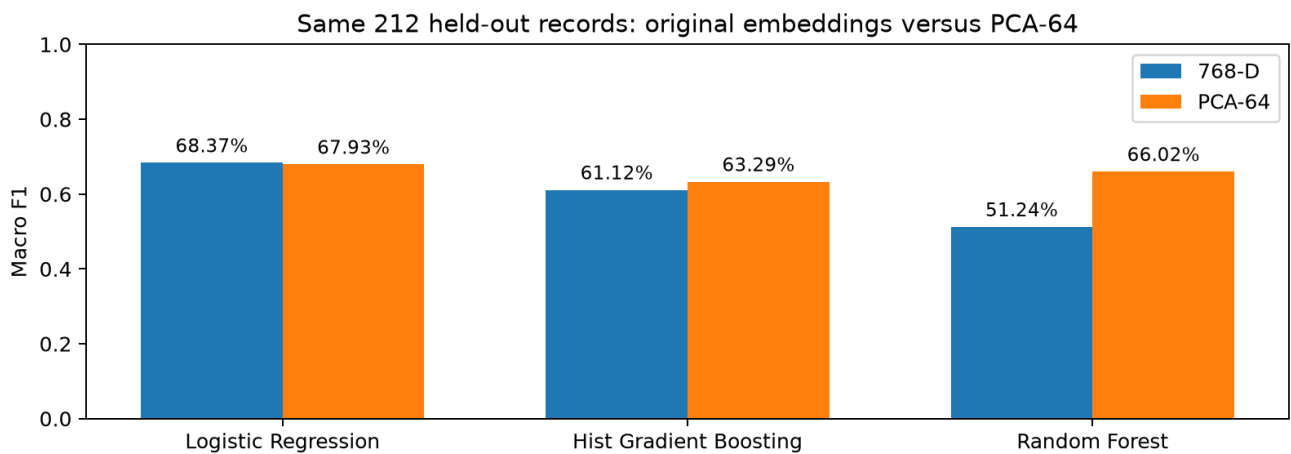
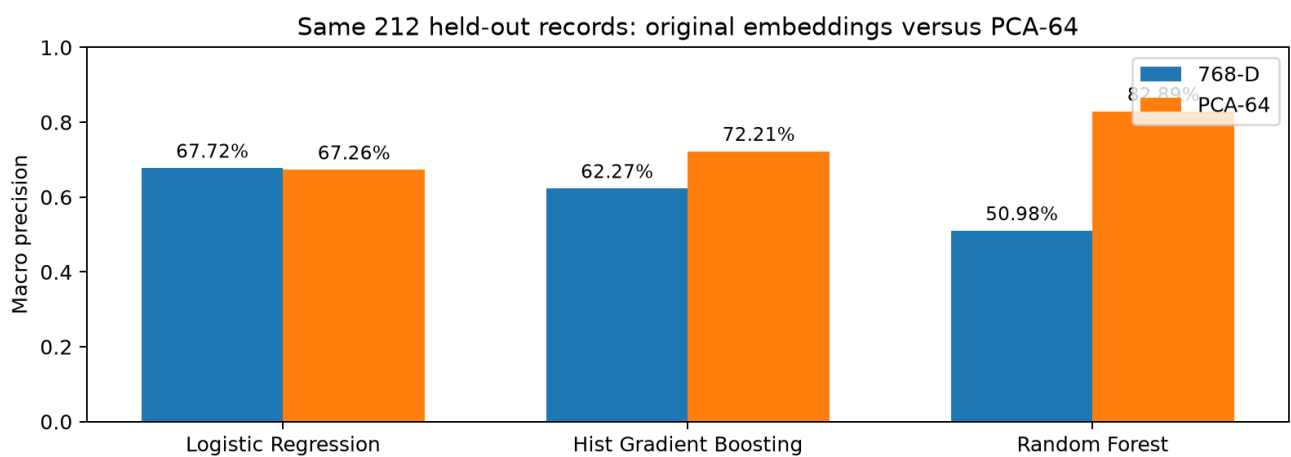
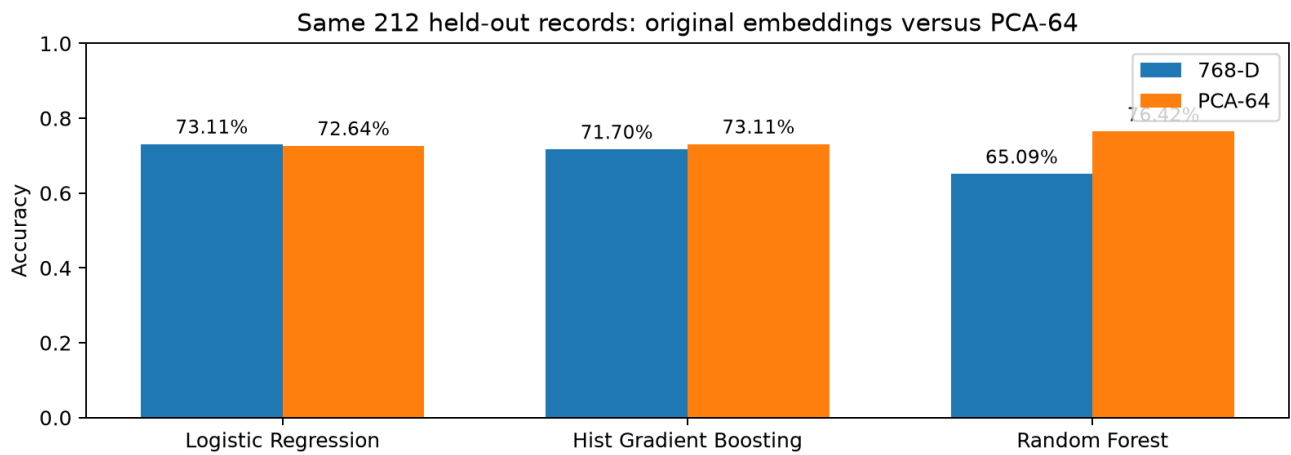
Text size	Classifier	Accuracy	Precision*	F1*
768-D	Logistic Regression	73.11%	67.72%	68.37%
768-D	Hist Gradient Boosting	71.70%	62.27%	61.12%
768-D	Random Forest	65.09%	50.98%	51.24%
PCA-64	Logistic Regression	72.64%	67.26%	67.93%
PCA-64	Hist Gradient Boosting	73.11%	72.21%	63.29%
PCA-64	Random Forest	76.42%	82.89%	66.02%

*Precision and F1 are macro-averaged. Scores describe this concept-based experiment; they are not the previous PDF's MPNet/English-translation experiment. Values are rounded for display; underlying CSV and JSON files retain full precision.

Text-only performance

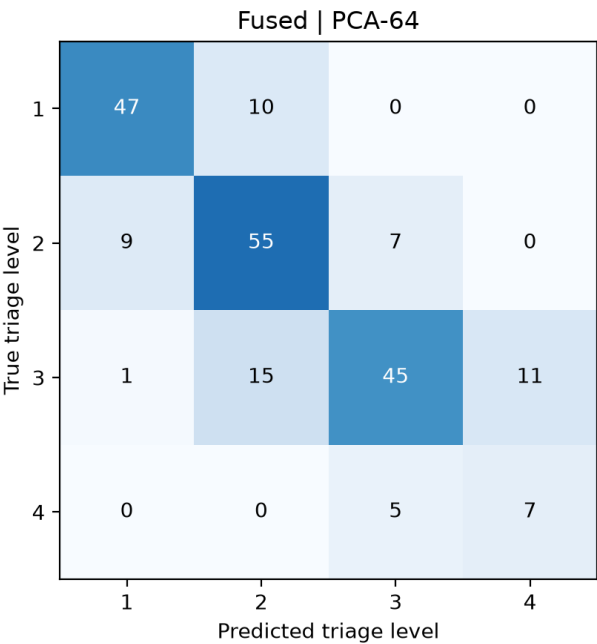
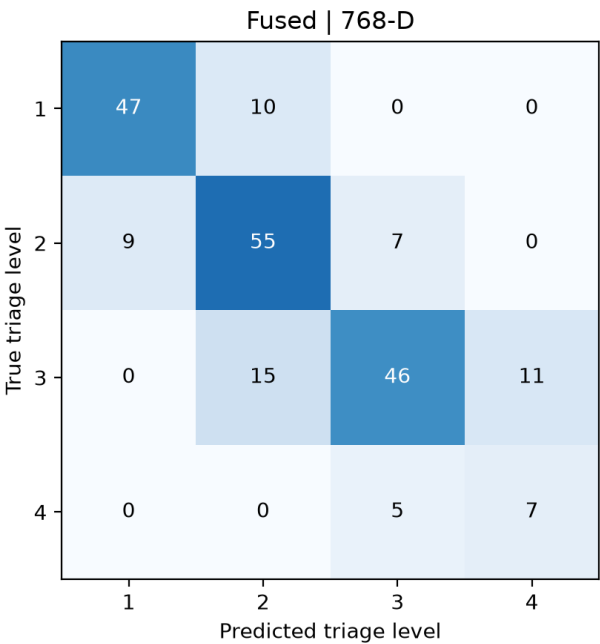
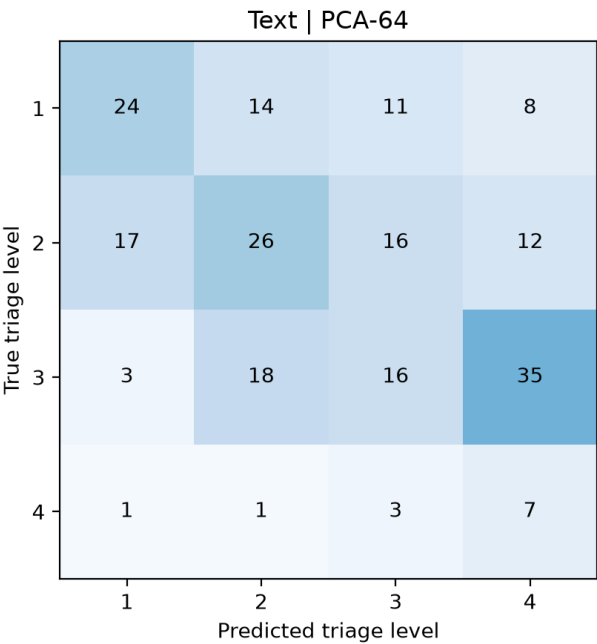
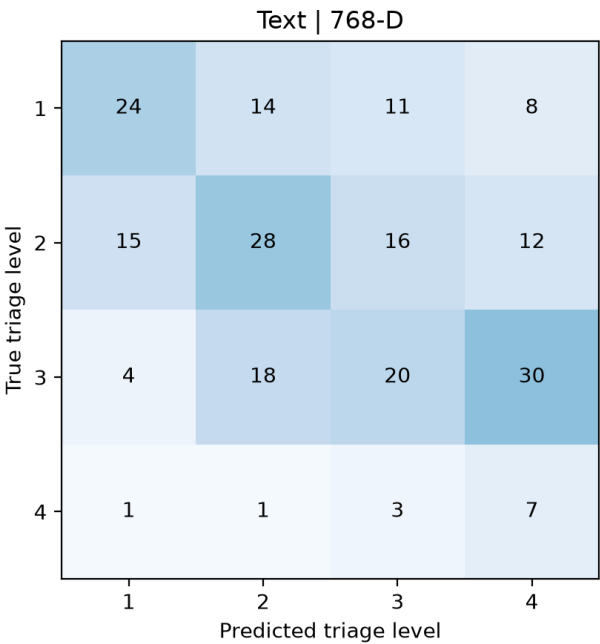


Performance with patient features



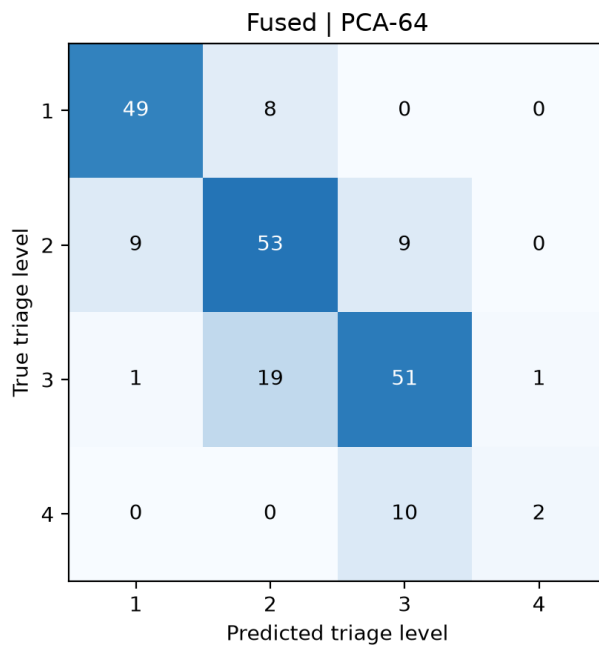
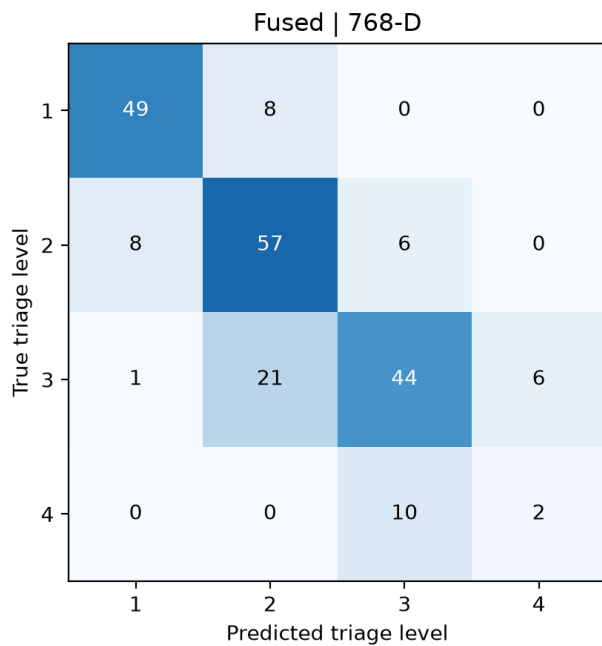
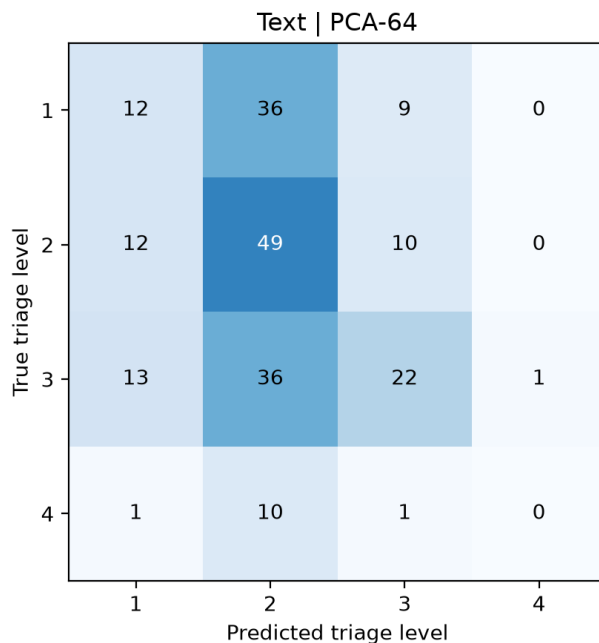
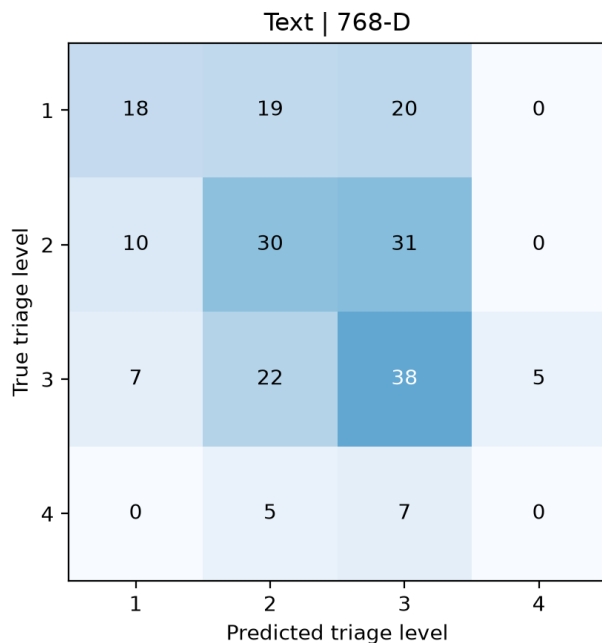
Logistic Regression: confusion matrices

Rows: true triage levels. Columns: predictions. Each matrix contains the same 212 test records. Top: text only. Bottom: text plus patient features. Left: 768-D; right: PCA-64.



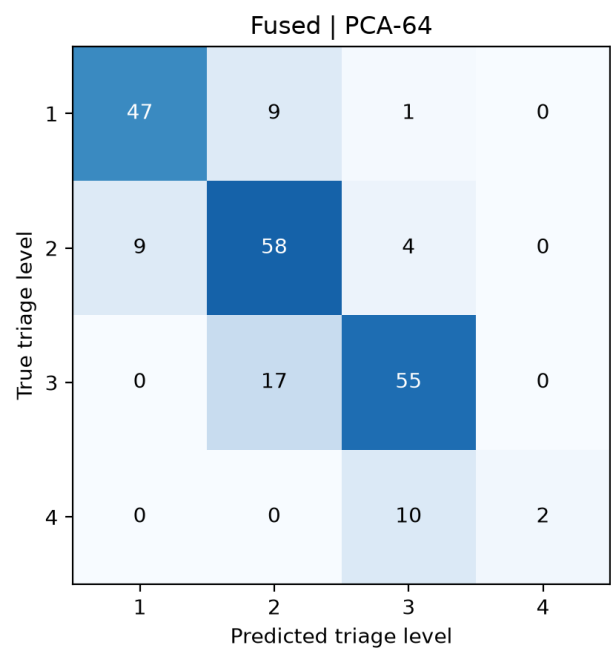
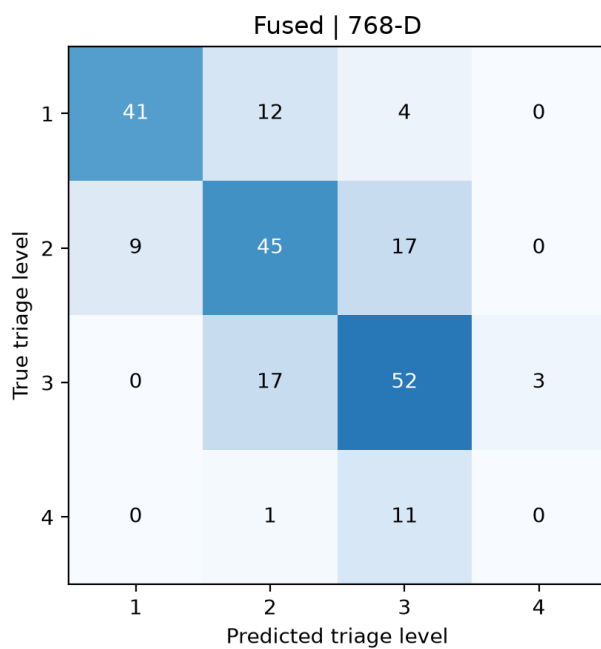
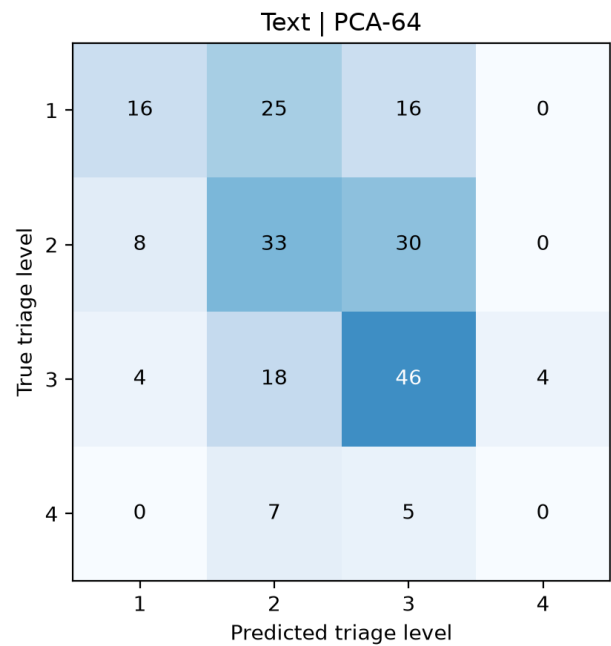
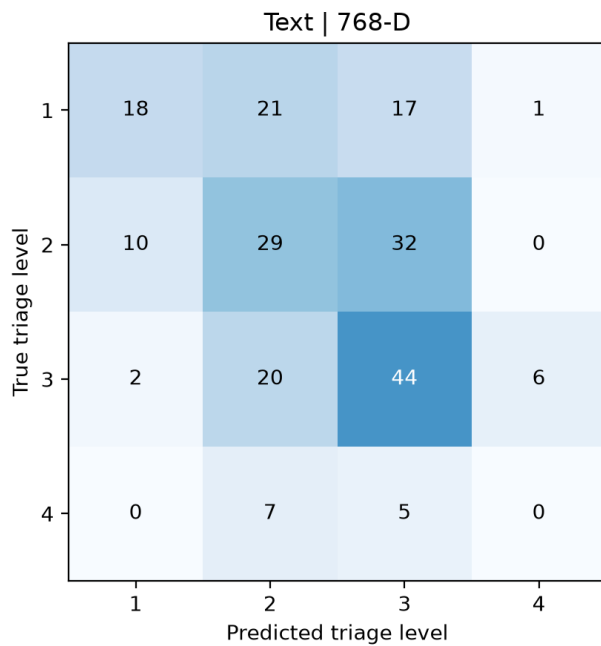
Hist Gradient Boosting: confusion matrices

Rows: true triage levels. Columns: predictions. Each matrix contains the same 212 test records. Top: text only. Bottom: text plus patient features. Left: 768-D; right: PCA-64.



Random Forest: confusion matrices

Rows: true triage levels. Columns: predictions. Each matrix contains the same 212 test records. Top: text only. Bottom: text plus patient features. Left: 768-D; right: PCA-64.



Patient-feature baseline and ordinal metrics

Patient features alone provide a reference for judging the contribution of concept text. These three models use the same 24 structured features and grouped holdout as the text comparisons.

Classifier	Accuracy	Precision*	Recall*	F1*
Logistic Regression	73.11%	67.92%	70.44%	68.60%
Hist Gradient Boosting	72.64%	63.85%	59.59%	59.71%
Random Forest	73.11%	74.04%	63.39%	65.54%

Ordinal metrics for all comparisons

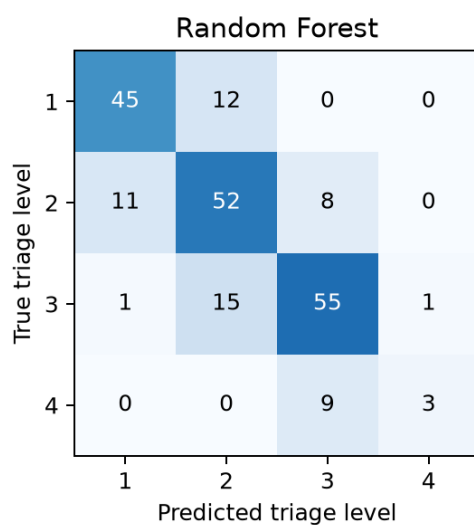
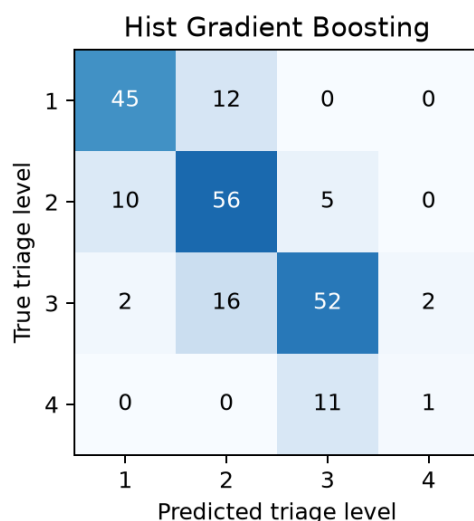
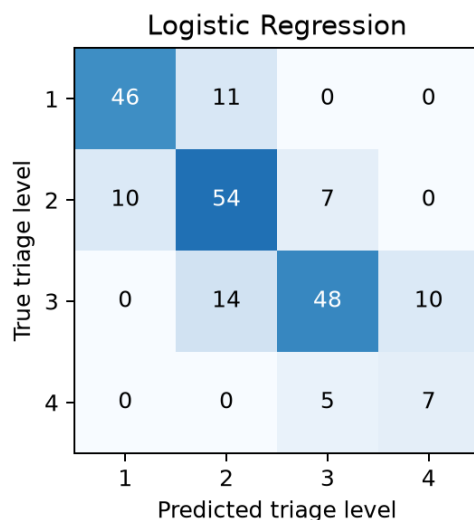
Lower triage numbers indicate greater urgency. Under-triage assigns a higher number than the true label; over-triage assigns a lower number. QWK is quadratic weighted kappa. MAE is the mean absolute error in triage levels.

Input / classifier	QWK	MAE	Under-triage	Over-triage
text_768+logreg	0.365	0.844	42.92%	19.81%
text_768+hgb	0.265	0.745	35.38%	24.06%
text_768+rf	0.316	0.703	36.32%	20.75%
fused_768+logreg	0.837	0.269	13.21%	13.68%
fused_768+hgb	0.803	0.288	9.43%	18.87%
fused_768+rf	0.711	0.373	16.98%	17.92%
text_64+logreg	0.379	0.868	45.28%	20.28%
text_64+hgb	0.100	0.769	26.42%	34.43%
text_64+rf	0.316	0.679	35.38%	19.81%
fused_64+logreg	0.826	0.278	13.21%	14.15%
fused_64+hgb	0.808	0.274	8.49%	18.40%
fused_64+rf	0.826	0.241	6.60%	16.98%
structured+logreg	0.836	0.269	13.21%	13.68%
structured+hgb	0.792	0.283	8.96%	18.40%
structured+rf	0.809	0.274	9.91%	16.98%

Key: text = concept embeddings; fused = embeddings plus patient features; structured = patient features only. logreg = Logistic Regression; hgb = Hist Gradient Boosting; rf = Random Forest.

Patient-feature baseline: confusion matrices

The same 212 test records are evaluated using patient features alone. Rows show true labels and columns show predictions.



Cross-validation and reproducibility

The uploaded script reports five-fold stratified cross-validation on the training matrices. The same procedure is used for added comparisons. PCA and scaling were fitted on the entire holdout-training partition before CV, and CV folds are not grouped by concept. These CV scores are exploratory; they are not leakage-free grouped validation.

Input / classifier	Folds	Mean macro F1	Standard deviation
text_768+logreg	5	36.35%	4.21%
text_768+hgb	5	30.53%	3.77%
text_768+rf	5	35.89%	6.50%
fused_768+logreg	5	74.56%	2.79%
fused_768+hgb	5	64.37%	4.38%
fused_768+rf	5	52.76%	7.93%
text_64+logreg	5	34.85%	2.98%
text_64+hgb	5	28.51%	3.71%
text_64+rf	5	36.35%	6.70%
fused_64+logreg	5	73.25%	2.39%
fused_64+hgb	5	66.33%	3.25%
fused_64+rf	5	66.46%	6.59%
structured+logreg	5	72.84%	3.24%
structured+hgb	5	71.15%	5.37%
structured+rf	5	74.93%	4.30%

Settings and scope

Logistic Regression: max_iter=2000, class_weight=balanced, random_state=42. Hist Gradient Boosting: max_iter=300, learning_rate=0.1, default unweighted loss, random_state=42. Added Random Forest: 300 trees, balanced class weights, random_state=42, n_jobs=2; remaining estimator settings use library defaults.

The uploaded script's six original PCA-64 / patient-feature results are reused from captured predictions. Additional 768-D and Random Forest comparisons are fitted separately. No deployed classifier is replaced. Exact source SHA-256: 5d3b9b5948d988533ab37d4d91d40b3303403aa9f47a467d2e0a021e9a9008ab.

Interpretation and Section 3.6

This is a single-split synthetic-data comparison, not clinical validation or an evaluation of live translation. Matching by exact concept string does not guarantee separation of all semantically similar complaints. Comparing many models on the holdout does not provide an untouched final validation set. Section 3.6 literature comparison remains pending because no verified comparator list and matched evaluation protocol were provided; no scores are invented.

Effect of dimensionality reduction and fusion

Differences below are percentage points. PCA change = PCA-64 minus 768-D. Fusion contribution = patient features plus concept text minus patient features alone. Positive values indicate a higher measured score on this holdout.

PCA-64 versus original 768-D

Inputs / classifier	Accuracy change	Macro F1 change
text / logreg	-2.83	-2.48
text / hgb	-1.42	-3.20
text / rf	+1.89	+0.65
fused / logreg	-0.47	-0.43
fused / hgb	+1.42	+2.17
fused / rf	+11.32	+14.78

Contribution of concept text to patient features

Text size / classifier	Accuracy change	Macro F1 change
768 / logreg	+0.00	-0.24
768 / hgb	-0.94	+1.42
768 / rf	-8.02	-14.31
64 / logreg	-0.47	-0.67
64 / hgb	+0.47	+3.59
64 / rf	+3.30	+0.47

These are observed differences from one split, without confidence intervals or significance testing. They should not be interpreted as established differences in clinical performance.

Text only: 768-D | class details

All values below are measured on the same 212 held-out records. Precision, recall and F1 are shown as percentages. Support is the number of true records in the class.

Logistic Regression

Class / average	Precision	Recall	F1	Support
1	54.55%	42.11%	47.52%	57
2	45.90%	39.44%	42.42%	71
3	40.00%	27.78%	32.79%	72
4	12.28%	58.33%	20.29%	12
macro avg	38.18%	41.91%	35.76%	212
weighted avg	44.32%	37.26%	39.27%	212

Accuracy: 37.26%

Hist Gradient Boosting

Class / average	Precision	Recall	F1	Support
1	51.43%	31.58%	39.13%	57
2	39.47%	42.25%	40.82%	71
3	39.58%	52.78%	45.24%	72
4	0.00%	0.00%	0.00%	12
macro avg	32.62%	31.65%	31.30%	212
weighted avg	40.49%	40.57%	39.55%	212

Accuracy: 40.57%

Random Forest

Class / average	Precision	Recall	F1	Support
1	60.00%	31.58%	41.38%	57
2	37.66%	40.85%	39.19%	71
3	44.90%	61.11%	51.76%	72
4	0.00%	0.00%	0.00%	12
macro avg	35.64%	33.38%	33.08%	212
weighted avg	43.99%	42.92%	41.83%	212

Accuracy: 42.92%

Text only: PCA-64 | class details

All values below are measured on the same 212 held-out records. Precision, recall and F1 are shown as percentages. Support is the number of true records in the class.

Logistic Regression

Class / average	Precision	Recall	F1	Support
1	53.33%	42.11%	47.06%	57
2	44.07%	36.62%	40.00%	71
3	34.78%	22.22%	27.12%	72
4	11.29%	58.33%	18.92%	12
macro avg	35.87%	39.82%	33.27%	212
weighted avg	41.55%	34.43%	36.33%	212

Accuracy: 34.43%

Hist Gradient Boosting

Class / average	Precision	Recall	F1	Support
1	31.58%	21.05%	25.26%	57
2	37.40%	69.01%	48.51%	71
3	52.38%	30.56%	38.60%	72
4	0.00%	0.00%	0.00%	12
macro avg	30.34%	30.16%	28.09%	212
weighted avg	38.81%	39.15%	36.15%	212

Accuracy: 39.15%

Random Forest

Class / average	Precision	Recall	F1	Support
1	57.14%	28.07%	37.65%	57
2	39.76%	46.48%	42.86%	71
3	47.42%	63.89%	54.44%	72
4	0.00%	0.00%	0.00%	12
macro avg	36.08%	34.61%	33.74%	212
weighted avg	44.79%	44.81%	42.96%	212

Accuracy: 44.81%

Patient features + text: 768-D | class details

All values below are measured on the same 212 held-out records. Precision, recall and F1 are shown as percentages. Support is the number of true records in the class.

Logistic Regression

Class / average	Precision	Recall	F1	Support
1	83.93%	82.46%	83.19%	57
2	68.75%	77.46%	72.85%	71
3	79.31%	63.89%	70.77%	72
4	38.89%	58.33%	46.67%	12
macro avg	67.72%	70.54%	68.37%	212
weighted avg	74.73%	73.11%	73.44%	212

Accuracy: 73.11%

Hist Gradient Boosting

Class / average	Precision	Recall	F1	Support
1	84.48%	85.96%	85.22%	57
2	66.28%	80.28%	72.61%	71
3	73.33%	61.11%	66.67%	72
4	25.00%	16.67%	20.00%	12
macro avg	62.27%	61.01%	61.12%	212
weighted avg	71.23%	71.70%	71.00%	212

Accuracy: 71.70%

Random Forest

Class / average	Precision	Recall	F1	Support
1	82.00%	71.93%	76.64%	57
2	60.00%	63.38%	61.64%	71
3	61.90%	72.22%	66.67%	72
4	0.00%	0.00%	0.00%	12
macro avg	50.98%	51.88%	51.24%	212
weighted avg	63.17%	65.09%	63.89%	212

Accuracy: 65.09%

Patient features + text: PCA-64 | class details

All values below are measured on the same 212 held-out records. Precision, recall and F1 are shown as percentages. Support is the number of true records in the class.

Logistic Regression

Class / average	Precision	Recall	F1	Support
1	82.46%	82.46%	82.46%	57
2	68.75%	77.46%	72.85%	71
3	78.95%	62.50%	69.77%	72
4	38.89%	58.33%	46.67%	12
macro avg	67.26%	70.19%	67.93%	212
weighted avg	74.21%	72.64%	72.90%	212

Accuracy: 72.64%

Hist Gradient Boosting

Class / average	Precision	Recall	F1	Support
1	83.05%	85.96%	84.48%	57
2	66.25%	74.65%	70.20%	71
3	72.86%	70.83%	71.83%	72
4	66.67%	16.67%	26.67%	12
macro avg	72.21%	62.03%	63.29%	212
weighted avg	73.03%	73.11%	72.13%	212

Accuracy: 73.11%

Random Forest

Class / average	Precision	Recall	F1	Support
1	83.93%	82.46%	83.19%	57
2	69.05%	81.69%	74.84%	71
3	78.57%	76.39%	77.46%	72
4	100.00%	16.67%	28.57%	12
macro avg	82.89%	64.30%	66.02%	212
weighted avg	78.04%	76.42%	75.36%	212

Accuracy: 76.42%

Patient features only | class details

All values below are measured on the same 212 held-out records. Precision, recall and F1 are shown as percentages. Support is the number of true records in the class.

Logistic Regression

Class / average	Precision	Recall	F1	Support
1	82.14%	80.70%	81.42%	57
2	68.35%	76.06%	72.00%	71
3	80.00%	66.67%	72.73%	72
4	41.18%	58.33%	48.28%	12
macro avg	67.92%	70.44%	68.60%	212
weighted avg	74.48%	73.11%	73.44%	212

Accuracy: 73.11%

Hist Gradient Boosting

Class / average	Precision	Recall	F1	Support
1	78.95%	78.95%	78.95%	57
2	66.67%	78.87%	72.26%	71
3	76.47%	72.22%	74.29%	72
4	33.33%	8.33%	13.33%	12
macro avg	63.85%	59.59%	59.71%	212
weighted avg	71.41%	72.64%	71.41%	212

Accuracy: 72.64%

Random Forest

Class / average	Precision	Recall	F1	Support
1	78.95%	78.95%	78.95%	57
2	65.82%	73.24%	69.33%	71
3	76.39%	76.39%	76.39%	72
4	75.00%	25.00%	37.50%	12
macro avg	74.04%	63.39%	65.54%	212
weighted avg	73.46%	73.11%	72.51%	212

Accuracy: 73.11%